Notes

[275]Exercise 12.2.6. Your definition should generalize the definition we gave for the case $n = 2$. The unit vector (u_1, u_2) will now have to be replaced by a unit vector $(u_1, u_2, u_3, \ldots, u_n)$.

[276]Exercise 12.3.3. Let

$$G(u,v,w) = \int_u^v f(x,w)\,dx.$$

Use Leibniz's rule, the fundamental theorem of calculus and an appropriate chain rule to obtain

$$F'(y) = \int_{u(y)}^{v(y)} \frac{\partial f}{\partial y}(x,y)\,dx$$

$$+f(u,y)\frac{dv}{dx} - f(u,y)\frac{du}{dx}.$$

[277]Exercise 12.4.3. See Exercises 11.2.6 and 11.2.7 for further discussion of these equivalent norms for $\mathbb{R}^n$.

[278]Exercise 12.4.5. This means that f is differentiable at x with respect to this definition if and only if it is differentiable at x with respect to Definition 12.16. Recall L is called *linear* if

$$L(\alpha x + \beta y) = \alpha L(x) + \beta L(y)$$

for all $\alpha, \beta \in \mathbb{R}$ and all $x, y \in \mathbb{R}^n$ and any linear function can be represented in the form

$$L(x) = \sum_{i=1}^{n} a_i x_i$$

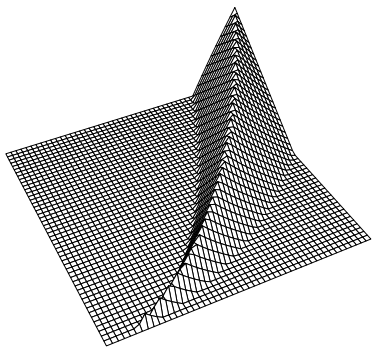
where

$$x = (x_1, \ldots, x_n).$$

Show that L must be as given in Definition 12.16, i.e., $a_i = f_i(x)$.

[279]Exercise 12.4.9. There are only two directions, $u_1 = 1$ and $u_1 = -1$. What are the two derivatives?

[280]Exercise 12.4.15. The picture illustrates a function that is mostly zero and has every directional derivative at $(0,0)$ zero but is not differentiable there.



Construct f to be continuous and such that

1. $f(x,y) = 0$ unless $x > 0$ and
$$x^2 < y < 3x^2,$$
2. for each $x > 0$, $f(x, 2x^2) = x$, and
3. $0 \le f(x,y) \le x$ for all (x,y) with $x > 0$.

Then all directional derivatives vanish at $(0,0)$ but when $k = 2h^2$, we obtain

$$\begin{aligned} h &= f_1(0,0)h + f_2(0,0)k + \varepsilon(h+k) \\ &= \varepsilon(h+2h^2) \end{aligned}$$

so

$$\varepsilon = h/(h+2h^2) \to 1$$

as $h \to 0$. This example also shows that differentiability is not a necessary condition for formula (29) in the text to be valid.

[281]Exercise 12.4.16. Assume that all but one of the partials are continuous at $x \in \mathbb{R}^n$ and the remaining partial is finite at x.

[282]Exercise 12.5.8. For (a): Would you expect

$$\frac{\partial A}{\partial x}$$

to be independent of whether it is y or P that is held fixed?

[283]Exercise 12.5.15. Use the result of Exercise 12.5.14.

[284]Exercise 12.5.16. The equations that transform (z,r) to (ρ,ϕ) are the same as those from (x,y) to (r,θ). The result is

$$\begin{aligned} &\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho}\frac{\partial^2 u}{\partial \phi^2} \\ &+ \frac{1}{\rho^2 \sin^2\phi}\frac{\partial^2 u}{\partial \theta^2} + \frac{2}{\rho}\frac{\partial u}{\partial \rho} + \frac{\cos\phi}{\rho^2}\frac{\partial u}{\partial \phi}. \end{aligned}$$

[285]Exercise 12.8.2. Calculate

$$A(x+h) - A(x) - A(h).$$

[286]Exercise 12.8.10. Make explicit the domains of the functions, the differentiability assumed, and the concluding chain.

[287]Exercise 12.8.13. Solve the equation

$$\begin{aligned} &\begin{pmatrix} \frac{\partial f}{\partial u} & \frac{\partial f}{\partial v} \\ \frac{\partial g}{\partial u} & \frac{\partial g}{\partial v} \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \text{ for } a,b,c \text{ and } d. \end{aligned}$$

Chapter 13

METRIC SPACES

13.1 Introduction

The real number system $\mathbb{R}$ is more than just a set of objects. It has various structures we were able to exploit in the preceding chapters. There was an algebraic structure that allowed us to add, subtract, multiply and divide numbers. There were order structures on $\mathbb{R}$ given by "$<$" (or "$\leq$", "$>$", "$\geq$"). And there was a notion of distance between pairs of members of $\mathbb{R}$ given by $|x-y|$. These structures interacted in various ways. In fact, we were able to characterize the real number system $\mathbb{R}$ via these structures: Speaking loosely, we can say that $\mathbb{R}$ is the only complete ordered field (Exercise 1.11.3). Some simple ways in which these structures interact can be found in the exercises at the end of this section.

Now, there are many sets of objects besides $\mathbb{R}$ that are important in analysis. We have encountered several in earlier chapters (e.g., sets of sequences or sets of functions). Many of these have natural algebraic structures (not necessarily that they form a field), natural order structures (not necessarily satisfying the axioms found in Section 1.3), and one or more natural notions of distance.

Consider, for example, the set $\mathcal{C}[0,1]$ of continuous functions on $[0,1]$. With the usual definitions for addition and multiplication of functions, $\mathcal{C}[0,1]$ is not a field. It does satisfy the axioms for a different algebraic structure, an *algebra* (see Exercise 13.1.2). If we write $f \leq g$ whenever $f(x) \leq g(x)$ for every $x \in (0,1)$ then "$\leq$" provides an order structure, but this structure does not satisfy the axioms indicated in Section 1.4 for $\mathbb{R}$. It does, however, satisfy another set of axioms, that for a *partially ordered set*. (Exercises 13.1.2 and 13.1.3 make these last two statements precise.)

There are also several notions of "distance" that can be considered between pairs of continuous functions. We shall see some of them in this chapter. We shall develop an abstract structure on sets of objects based on the distance concept. The resulting structure is called a *metric space*. This abstract structure can be imposed on any nonempty set. It is most efficient to develop the material in an abstract fashion, but we shall spend a good deal of time applying the theory to specific "concrete" metric spaces. Our concrete examples will be quite varied. In particular, our applications will include metric spaces whose elements consist of such diverse objects as sequences, functions and sets.

You may wish to review how the notions of distance between pairs of points in $\mathbb{R}$ (and in $\mathbb{R}^n$) figured in the preceding chapters. From the distance notion, we obtained other central concepts of elementary analysis (open set, sequence limits, neighborhood,

limits, continuous functions, etc.). The entire subject of elementary real analysis in $\mathbb{R}$ relies fundamentally on the distance notion.

Exercises

13.1.1 Verify each of the following facts (or find them explicitly stated in earlier sections of the book). Each of these facts provides an example of ways in which the algebraic, order, and/or distance structures interact. In each case, indicate which structures appear in the statement.

(a) $|x-z| \leq |x-y| + |y-z|$ for all $x, y, z \in \mathbb{R}$.
(b) If $\{a_k\} \to a$ and $\{b_k\} \to b$, then $\{a_k + b_k\} \to a + b$.
(c) If $x > 0$ and $y > 0$, then there exists $n \in \mathbb{N}$ such that $nx > y$.
(d) For every $n \in \mathbb{N}$, $n^2 \geq n$.
(e) If $\{a_k\} \to a$ and $a_k \leq b_k \leq a$ for all $k \in \mathbb{N}$, then $b_k \to a$.
(f) Every nonempty bounded set $E \subset \mathbb{R}$ has a least upper bound.

13.1.2 Let $\mathcal{C}[0,1]$ denote the continuous functions on $[0,1]$ furnished with the usual notions of addition and multiplication of functions. Prove that $\mathcal{C}[0,1]$ is an *algebra* of functions, that is, that it satisfies the following axioms:

(a) $\mathcal{C}[0,1]$ is a vector space.
(b) $f(gh) = (fg)h$ for all $f, g, h \in \mathcal{C}[0,1]$.
(c) $f(g+h) = fg + fh$ for all $f, g, h \in \mathcal{C}[0,1]$.
(d) $(cf)g = c(fg)$ for all $c \in \mathbb{R}$ and $f, g \in \mathcal{C}[0,1]$.

SEE NOTE 288

13.1.3 Let S be a set. A relation $a \preceq b$ defined for certain pairs in S is called a *partial order* on S if it satisfies the following axioms.

(a) $a \preceq a$ for all $a \in S$.
(b) If $a \preceq b$ and $b \preceq a$, then $a = b$.
(c) If $a \preceq b$ and $b \preceq c$, then $a \preceq c$.

We then say that S *is partially ordered by* $\preceq$. Show that the set $\mathcal{C}[0,1]$ of continuous functions on $[0,1]$ is partially ordered by the relation $f \preceq g$ if $f(x) \leq g(x)$ for all $x \in [0,1]$.

13.1.4 Find a partial order on the set $\mathcal{C}[0,1]$ of continuous functions on $[0,1]$ that is different from the partial order in Exercise 13.1.3.

SEE NOTE 289

13.1.5 Define a relation $f \preceq g$ on the set $\mathcal{C}[0,1]$ of continuous functions on $[0,1]$ if

$$\int_0^1 f(t)\,dt \leq \int_0^1 g(t)\,dt.$$

Is this a partial order as defined in Exercise 13.1.3?

13.1.6 Let $\mathcal{S}$ denote the set of all subsets of $\mathbb{R}$. Show that $\mathcal{S}$ is partially ordered by "$\subset$".

13.1.7 Let $\mathcal{P}$ denote the polynomials defined on $\mathbb{R}$. For $p_1, p_2 \in \mathcal{P}$, write $p_2 \succ p_1$ if there exists $n \in \mathbb{N}$ such that $p_2(x) > p_1(x)$ for all $x \geq n$. Does "$\succ$" satisfy the order conditions that are given for "$<$" (relative to $\mathbb{R}$) in Section 1.4? What would your answer be if we replaced $\mathcal{P}$ by the set of continuous functions on $\mathbb{R}$?

13.2 Metric Spaces—Specific Examples

In Section 1.10 we introduced the metric structure of the real line $\mathbb{R}$ via the concept of absolute value. For $x, y \in \mathbb{R}$, we called

$$d(x,y) = |x-y|$$

the distance between x and y. We identified four properties that the distance function d possesses:

1. $d(x,y) \geq 0$

   [All distances are positive or zero.]

2. $d(x,y) = 0$ if and only if $x = y$

   [Different points are at a positive distance apart.]

3. $d(x,y) = d(y,x)$

   [Distance is symmetric, that is, the distance from x to y is the same as that from y to x.]

4. $d(x,y) \leq d(x,z) + d(z,y)$

   [The triangle inequality; it is no farther to go directly from x to y than to go from x to z and then to y.]

All of the work we did connected with the limit concept and convergence rested ultimately on these four properties of d. Readers who worked through Chapter 11 and 12 on the euclidean spaces $\mathbb{R}^n$ have seen that the euclidean distance between points x and y,

$$d(x,y) = \|x-y\|,$$

obeyed the same properties.

With these facts in mind, we shall base our definition of metric space on a *metric* that possesses these properties. The following concepts are due to Maurice René Fréchet (1878–1973), who introduced them in his 1906 dissertation. The name "metric space" was not originally used but was introduced later in 1914, by Felix Hausdorff (1869–1942), who took Fréchet's ideas, built on them, and extended them to create a branch of mathematics now known as topology. Metric spaces play a fundamental role in the study of topology.

Definition 13.1: Let X be any nonempty set. A function

$$d : X \times X \to \mathbb{R}$$

is called a *metric* if it satisfies the following four conditions:

1. $d(x,y) \geq 0$ for all $x, y \in X$.
2. $d(x,y) = 0$ if and only if $x = y$.
3. $d(x,y) = d(y,x)$ for all $x, y \in X$.

4. $d(x,y) \leq d(x,z) + d(z,y)$ for all $x,y,z \in X$.

Example 13.2: We know already that on the real numbers $\mathbb{R}$ the function

$$d(x,y) = |x-y|$$

is a metric since it is our model on which we based the definition. This is frequently called the *usual metric* since in most (but not all) studies of real numbers this is the metric that is used. For a collection of interesting examples we can consider any subset $A \subset \mathbb{R}$. The metric is the same: More precisely the metric we take on A is the function d_A defined on $A \times A$ by $d_A(x,y) = |x-y|$. Usually this level of precision is not needed and we just refer to the "usual metric." Each of the following examples will prove useful in the sequel:

1. $A = \mathbb{Q}$, the set of rational numbers
2. $A = \mathbb{N}$, the set of natural numbers
3. $A = K$, the Cantor set
4. $A = \{1/k : k = 1,2,3,\dots\}$

◀

Example 13.3: (Discrete Space) Let X be any nonempty set. For points x and y in X let

$$d(x,y) = 1 \text{ if } x \neq y \ \ (d(x,x) = 0).$$

Then d is a metric on X. (Check this.) This space may not appear to be of any interest, but it is useful to keep in mind because it can be helpful in avoiding certain misconceptions. (See Example 13.18.) ◀

Example 13.4: (The Euclidean Plane) Let

$$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R} = \{(x_1,x_2) : x_1,x_2 \in \mathbb{R}\}.$$

For $x = (x_1,x_2)$ and $y = (y_1,y_2)$ define

$$d_2(x,y) = \sqrt{(x_1-y_1)^2 + (x_2-y_2)^2}.$$

You will recognize that this is the usual way in elementary geometry that distances in the plane are computed and can readily verify that d_2 is indeed a metric on $\mathbb{R}^2$. Condition (4) is merely the familiar triangle inequality. It states intuitively that the "shortest distance between two points" is along the line joining those points. A rigorous proof depends upon the Cauchy-Schwarz inequality, with which you may be familiar from the exercises (e.g., Exercises 3.5.13, 3.5.14, and 8.3.10). This material was also covered in Chapter 11. ◀

Example 13.5: (Euclidean n-dimensional space) Let

$$\mathbb{R}^n = \mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}$$
$$= \{(x_1, x_2, \ldots, x_n) : x_i \in \mathbb{R},\ i = 1, 2, \ldots, n\}.$$

For $x = (x_1, x_2, \ldots, x_n)$ and $y = (y_1, y_2, \ldots, y_n)$, define

$$d_2(x,y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \cdots + (x_n - y_n)^2}.$$

Once again, d_2 is a metric on $\mathbb{R}^n$. The triangle inequality (4) follows from the Cauchy-Schwarz inequality. (See Section 11.2.) ◀

Metric Spaces When d is a metric on a set X we refer to X as a metric space furnished with the metric d. The precise language is in the definition, but you will likely find that lecturers have less formal ways of talking about metric spaces.

Definition 13.6: A *metric space* is an ordered pair (X, d), where X is a nonempty set and d is a metric on X.

Sometimes the metric on X is understood from the context. In this case we often shorten our notation by calling the metric space "X" instead of "(X, d)." In particular, when no confusion can arise, we assume that $\mathbb{R}$ is furnished with the usual euclidean metric

$$d(x,y) = |x - y|.$$

There can be many different metrics on a set X. If more than one metric is under consideration, we must make it clear which metric is considered at a particular point of the discussion. (See Exercise 13.2.6 for two further interesting metrics on the space $\mathbb{R}^2$. Exercise 13.6.31 illustrates how a sloppy treatment when two metrics are being used can lead to errors.)

We shall discuss a few additional metric spaces in the next section.

Exercises

13.2.1 Which of the following functions defined for pairs of numbers x and y are metrics on $\mathbb{R}$?

(a) $d(x,y) = |x| + |y|$
(b) $d(x,y) = (x - y)^2$
(c) $d(x,y) = \sqrt{|x - y|}$
(d) $d(x,y) = \min\{1, |x - y|\}$
(e) $d(x,y) = \frac{|x-y|}{1+|x-y|}$
(f) $d(x,y) = 1$ if $x \neq y$ and $d(x,y) = 0$ if $x = y$

SEE NOTE 290

13.2.2 Let $\mathbb{R}^+$ denote the set of positive real numbers. Show that

$$d(x,y) = \left| \frac{1}{x} - \frac{1}{y} \right|$$

is a metric on $\mathbb{R}^+$.

13.2.3 Let $\{x_1, x_2, \ldots, x_m\}$ be a finite set of points in a metric space (X, d). Show that

$$d(x_1, x_m) \leq \sum_{i=1}^{m-1} d(x_i, x_{i+1}).$$

13.2.4 Show that a function

$$d : X \times X \to \mathbb{R}$$

is a metric if and only if it satisfies the following three conditions:

(a) $d(x,y) \geq 0$ for all $x, y \in X$

(b) $d(x,y) = 0$ if and only if $x = y$

(c) $d(x,y) \leq d(z,x) + d(z,y)$ for all $x, y, z \in X$

As a question of mathematical taste, would you prefer to use these conditions rather than the four in Definition 13.1 for the definition of this term?

13.2.5 Let $X = \{x_1, x_2, x_3, \ldots, x_n\}$ be a finite set and let d be a metric on X. Consider the $n \times n$ matrix whose i,j entry is $d(x_i, x_j)$. What properties must such a matrix have?

13.2.6 Let $X = \mathbb{R}^2$. For $x = (x_1, x_2)$, $y = (y_1, y_2)$ let

$$\begin{aligned} d_1(x,y) &= |x_1 - y_1| + |x_2 - y_2|, \\ d_\infty(x,y) &= \max\{|x_1 - y_1|, |x_2 - y_2|\}. \end{aligned}$$

Show that d_1 and d_∞ are metrics on X. Show that these are distinct from each other and from the metric d_2 used in Example 13.4 (which is the most commonly used metric in $\mathbb{R}^2$).

13.2.7 Let (X, d) be a metric space. Define a function $e : X \times X \to \mathbb{R}$ by

$$e(x,y) = \frac{d(x,y)}{1 + d(x,y)}.$$

Prove that e is a metric, that $e(x,y) \leq d(x,y)$, and that $e(x,y) \leq 1$ for all $x, y \in X$.
SEE NOTE 291

13.2.8 Let (X, d) be a metric space. Define a function $e : X \times X \to \mathbb{R}$ by

$$e(x,y) = \min\{1, d(x,y)\}.$$

Prove that e is a metric and that $e(x,y) \leq 1$ for all $x, y \in X$.

13.2.9 **(Product Spaces)** Given two metric spaces we can form a product metric space. Let (X_1, d_1) and (X_2, d_2) be metric spaces. The set

$$X_1 \times X_2 = \{(x_1, x_2) : x_1 \in X_1, x_2 \in X_2\}$$

is called the product or Cartesian product of X_1 and X_2. For

$$u = (x_1, x_2) \text{ and } v = (y_1, y_2) \text{ in } nX_1 \times X_2$$

define

$$d(u,v) = d_1(x_1, y_1) + d_2(x_2, y_2).$$

(a) Prove that d is a metric on $X_1 \times X_2$. (It is called the product metric.)

(b) Let $X_1, X_2 = \mathbb{R}$, $d_1 = d_2$ be the usual euclidean metric on $\mathbb{R}$. Calculate $d(u,v)$, where $u = (0,1)$ and $v = (-3,4)$ are points in $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$.

13.2.10 **(Normed Vector Spaces)** Let X be a vector space. A *norm* on X is a real-valued function on X so that if x, $y \in X$ and $\alpha \in \mathbb{R}$, then

(i) $\|x\| \geq 0$, and $\|x\| = 0$ if and only if x is the zero of X,

(ii) $\|\alpha x\| = |\alpha|\,\|x\|$, and

(iii) $\|x+y\| \leq \|x\| + \|y\|$.

Show that a vector space equipped with a norm can be considered a metric space by defining $d(x,y) = \|x-y\|$.

13.3 Additional Examples

We have already mentioned that the elements of a metric space can be objects of various sets. In this section we introduce several important metric spaces whose elements are sequences or functions. Exercise 13.3.9 provides an example of an important metric space whose elements are sets—the closed nonempty subsets of $[0,1]$.

13.3.1 Sequence Spaces

Example 13.7: (Hilbert Space—ℓ_2) An immediate generalization of euclidean n-dimensional space $\mathbb{R}^n$ arises when we replace n-tuples

$$x = (x_1, x_2, \ldots, x_n)$$

with sequences

$$x = \{x_1, x_2, \ldots\}.$$

Let ℓ_2 denote the set of all sequences of real numbers such that

$$\sum_1^\infty (x_k)^2 < \infty.$$

For

$$x = \{x_1, x_2, \ldots\} \text{ and } y = \{y_1, y_2, \ldots\}$$

define

$$d_2(x,y) = \sqrt{\sum_{k=1}^\infty (x_k - y_k)^2}.$$

The resulting metric space ℓ_2 is usually called classical (sequential) Hilbert space, named after one of the most important and influential mathematicians of his time, David Hilbert (1862–1943). Let us verify that d_2 is a metric on ℓ_2. It is clear that for $x, y \in \ell_2$, $d_2(x,y) \geq 0$, but we must check that $d_2(x,y)$ is finite. From the elementary inequality

$$(x_k \pm y_k)^2 \leq 2(x_k^2 + y_k^2)$$

we see that the convergence of the series $\sum_1^\infty x_k^2$ and $\sum_1^\infty y_k^2$ implies the convergence of the series $\sum_1^\infty (x_k - y_k)^2$. Thus $d_2(x,y) < \infty$ for all $x, y \in \ell_2$. It is clear that $d_2(x,y) = 0$ if and only if $x = y$. The symmetry condition (3) is also apparent.

Let us verify the triangle inequality. In this space, it takes the form

$$\sqrt{\sum_{k=1}^\infty (x_k - y_k)^2} \leq \sqrt{\sum_{k=1}^\infty (x_k - z_k)^2} + \sqrt{\sum_{k=1}^\infty (z_k - y_k)^2}. \tag{1}$$

From the Cauchy-Schwarz inequality we can deduce that for all $n \in \mathbb{N}$,

$$\sqrt{\sum_{k=1}^{n}(x_k-y_k)^2} \le \sqrt{\sum_{k=1}^{n}(x_k-z_k)^2} + \sqrt{\sum_{k=1}^{n}(z_k-y_k)^2}. \tag{2}$$

We have left the deduction of (2) from the Cauchy-Schwarz inequality as an exercise. Letting $n \to \infty$ in (2), we obtain (1), verifying the triangle inequality for d_2. ◀

Note. We mention that Example 13.7 is known as *real* ℓ_2. The complex version, which is often studied, is applied to sequences of complex numbers $\{z_1, z_2, \dots\}$ satisfying $\sum_1^\infty |z_k|^2 < \infty$ and is the same except that here we must define

$$d_2(z,w) = \sqrt{\sum_{1}^{\infty}|z_k-w_k|^2}$$

for a pair of sequences of complex numbers,

$$z = \{z_1, z_2, \dots\} \text{ and } w = \{w_1, w_2, \dots\}.$$

The proof that this is again a metric is similar, using now properties of the complex modulus.

Example 13.8: Now let ℓ_1 be the set of sequences $\{x_k\}$ of real numbers for which $\sum_1^\infty |x_k| < \infty$, and let

$$d_1(x,y) = \sum_{1}^{\infty}|x_k-y_k|.$$

We verify easily that d_1 is a metric on ℓ_1. The metric space (ℓ_1, d_1) can be described loosely as the space of all absolutely convergent series. ◀

Example 13.9: Let ℓ_∞ denote the set of all bounded sequences of real numbers, let $x = \{x_1, x_2, \dots\}$ and $y = \{y_1, y_2, \dots\}$, and let

$$d_\infty(x,y) = \sup_k |x_k - y_k|.$$

Once again, it is easy to verify that (ℓ_∞, d_∞) is a metric space. (We leave this verification as Exercise 13.3.3.) This space ℓ_∞ (sometimes denoted by m) can be described loosely as the space of all bounded sequences. ◀

These three spaces ℓ_1, ℓ_2, and ℓ_∞ are related in a number of ways. For example, you should be able to use your study of series convergence to prove that

$$\ell_1 \subset \ell_2 \subset \ell_\infty.$$

Thus, on the smallest of these spaces ℓ_1, we can define three metrics; all are different and all are important. For example, if

$$x = \{3,4,0,0,\dots\} \quad \text{and} \quad y = \{0,0,\dots\},$$

then $d_1(x,y) = 7$, $d_2(x,y) = 5$ and $d_\infty(x,y) = 4$. Note the inequalities

$$d_1(x,y) \ge d_2(x,y) \ge d_\infty(x,y)$$

for this example. Do you think these inequalities are valid in general whenever x and y are *any* absolutely convergent sequences?

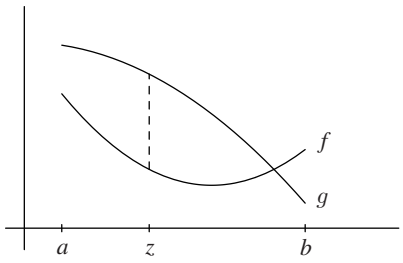


Figure 13.1. Distance between functions f and g in $\mathcal{C}[a,b]$.

13.3.2 Function Spaces

We often use the term "function space" when the elements of a metric space are functions. Function spaces have played a major role in the development of twentieth-century analysis.

Example 13.10: We use $\mathcal{C}[a,b]$ to denote the set of continuous functions on $[a,b]$. For a "distance" between two continuous functions f and g we measure the largest vertical distance between points on their graphs. Thus we shall employ the metric

$$d(f,g) = \max_{a\le t\le b} |f(t)-g(t)|.$$

Figure 13.1 illustrates a point z in the interval at which the maximum is attained. (We know there must be a maximum since $|f-g|$ is continuous.) It is easy to verify that d is a metric on $\mathcal{C}[a,b]$. ◀

Example 13.11: The space of continuous functions on $[a,b]$ can be enlarged by passing to the collection of all *bounded* functions on $[a,b]$. We use $M[a,b]$ to denote the set of bounded functions defined on $[a,b]$. For $f,g \in M[a,b]$ we define a metric in much the same way by measuring the largest vertical distance between the graphs of f and g, but this time we cannot be assured that a maximum is attained. Thus we write

$$d(f,g) = \sup_{a\le t\le b} |f(t)-g(t)|. \tag{3}$$

This is often called the *sup metric*. It is easy to verify that $(M[a,b],d)$ is a metric space. Note especially that if f and g are continuous functions, then the distance $d(f,g)$ assigned by this metric is the same as the distance assigned by the metric in the space $\mathcal{C}[a,b]$ of the preceding example. ◀

Subspaces These last two examples introduce the important concept of "subspace" of a metric space. The space of bounded functions $M[a,b]$ contains inside it the space of continuous functions $\mathcal{C}[a,b]$ and the metrics agree.

(Since our subject matter is metric spaces, we intend a subspace to be understood relative to that structure. You may also have studied "subspaces" in the context of vector spaces; this has an entirely different meaning and you should avoid confusing the two meanings.)

Definition 13.12: If (X,d) is a metric space and $A \subset X$, then (A,d) is called a *subspace* of (X,d).

Note that the metric must be the same for A and for X.

Example 13.13: For example, with $X = \mathbb{R}$ and the usual metric

$$d(x,y) = |x-y|,$$

any nonempty subset A of $\mathbb{R}$ becomes a subspace of (X,d) as long as we use the same metric $d(x,y) = |x-y|$ for $x,y \in A$ on the subspace. ◀

Example 13.14: Let X denote the continuous functions on $[a,b]$. Define

$$e(f,g) = \int_a^b |f(t) - g(t)|\,dt.$$

Again we verify easily that (X,e) is a metric space. Note that if $f(t) = t$ and $g(t) = t^2$ on $[0,1]$, then

$$e(f,g) = \int_0^1 |t - t^2|\,dt = \frac{1}{6}.$$

The metric from Example 13.10 is quite different since it would give

$$d(f,g) = \max_{0 \le t \le 1} |t - t^2| = 1/4.$$

Thus while X is contained inside the set $M[a,b]$ it cannot be considered a subspace of the space $M[a,b]$ since it has been equipped with a different metric. ◀

Exercises

13.3.1 Deduce for all $n \in \mathbb{N}$, that

$$\sqrt{\sum_{k=1}^{n} (x_k - y_k)^2} \le \sqrt{\sum_{k=1}^{n} (x_k - z_k)^2} + \sqrt{\sum_{k=1}^{n} (z_k - y_k)^2} \tag{4}$$

from the Cauchy-Schwarz inequality.

13.3.2 Let ℓ_1 be the set of sequences $\{x_k\}$ of real numbers for which $\sum_1^\infty |x_k| < \infty$, and let

$$d_1(x,y) = \sum_1^\infty |x_k - y_k|.$$

Show that d_1 is a metric on ℓ_1.

13.3.3 Let ℓ_∞ denote the set of all bounded sequences of real numbers, let

$$x = \{x_1, x_2, \dots\} \text{ and } y = \{y_1, y_2, \dots\}$$

belong to ℓ_∞, and let

$$d_\infty(x,y) = \sup_k |x_k - y_k|.$$

Show that d_∞ is a metric on ℓ_∞.

13.3.4 Let $\mathcal{C}[0,1]$ consist of the continuous functions on $[0,1]$ furnished with the metric

$$d(f,g) = \int_0^1 |f(t) - g(t)|\,dt.$$

Show that d is a metric on $\mathcal{C}[0,1]$ different from the one in Example 13.10.

13.3.5 Let $\mathcal{R}[0,1]$ consist of all integrable functions on $[0,1]$ (not necessarily continuous). Let

$$d(f,g) = \int_0^1 |f(t) - g(t)|\,dt.$$

Show that d is *not* a metric on $\mathcal{R}[0,1]$.

SEE NOTE 292

13.3.6 Verify the inclusions

$$\ell_1 \subset \ell_2 \subset \ell_\infty$$

for the sequence spaces in this section. Is any one of these a subspace of any other?

SEE NOTE 293

13.3.7 Let $M(\mathbb{R})$ denote the collection of all bounded real-valued functions on $\mathbb{R}$ and let

$$d(f,g) = \sup\{|f(t) - g(t)| : t \in \mathbb{R}\}.$$

Show that d is a metric on $M(\mathbb{R})$. Which of the following are subspaces of $M(\mathbb{R})$?

$\mathcal{A}$ = the constant functions on $\mathbb{R}$

$\mathcal{P}$ = the polynomials

$\mathcal{C}$ = the continuous functions

$\mathcal{S}$ = the set of functions f of the form $f(t) = a\sin(nt) + b\cos(nt)$ for a, $b \in \mathbb{R}$, $n \in \mathbb{N}$

SEE NOTE 294

13.3.8 Let $\mathcal{K}$ consist of the nonempty closed subsets of $[0,1]$. For $A, B \in \mathcal{K}$ let

$$\text{dist}(A,B) = \inf\{|a-b| : a \in A, b \in B\}.$$

This is often called the "distance between A and B." Show that dist is *not* a metric on $\mathcal{K}$.

13.3.9 **(The Hausdorff Metric)** Since the "distance function" in Exercise 13.3.8 is not a true metric, let us define a metric on $\mathcal{K}$, the family of nonempty closed subsets of $[0,1]$, that captures the idea that the distance between two sets A and B in $\mathcal{K}$ is less than δ if every point of A is within δ of some point of B, and vice versa. For $A \in \mathcal{K}$ and $\delta > 0$, let A_δ be the union of all closed intervals of length 2δ centered at points of A. Define d by

$$d(A,B) = \inf\{\delta > 0 : A \subset B_\delta \text{ and } B \subset A_\delta\} \qquad (*).$$

(a) Verify that $d(A,B)$ measures the greatest distance that a point in A can be from the set B or a point in B can be from the set A.

(b) Show that d is a metric on $\mathcal{K}$. It is called the *Hausdorff metric* on the space of (nonempty) closed subsets of $[0,1]$.

(c) Let $A = \{1/n : n \in \mathbb{N}\} \cup \{0\}$, $B = [0,1]$ and $C = \{1/2\}$. Calculate $d(A,B)$, $d(B,C)$ and $d(A,C)$.

(d) If we replace the family $\mathcal{K}$ with the family of *all* nonempty subsets of $[0,1]$ and define d by $(*)$, we would not get a metric. Which of the conditions for a metric would fail? What would be the value of $d(A,B)$ if $A = \mathbb{Q} \cap [0,1]$ and $B = [0,1]$?

13.3.10 Let X be the set of continuous functions on $(0,1)$. For $x, y \in X$ let $U(x,y) = \{t \in (0,1) : x(t) \neq y(t)\}$. Then $U(x,y)$ is a disjoint union of intervals. Let $d(x,y)$ be the sum of the lengths of these intervals. Verify that (X,d) is a metric space.

13.4 Convergence

We recall that a sequence $\{x_k\}$ of real numbers converges to x_0 if and only if $|x_k - x_0|$ converges to zero, that is, if and only if

$$d(x_k, x_0) \to 0,$$

where d is the usual metric. This meaning of convergence carries over to any metric space.

Definition 13.15: Let (X,d) be a metric space. Let $\{x_k\}$ be a sequence of members of X and let $x_0 \in X$. If $\lim_{k\to\infty} d(x_k, x_0) = 0$ we say $\{x_k\}$ *converges to* x_0 and we write $\lim_{k\to\infty} x_k = x_0$ or $x_k \to x_0$.

It is clear from the way that sequence convergence in a metric space has been defined that convergence in $\mathbb{R}$ (with the usual metric) is precisely the notion of convergence as we have studied it so far. What sequences will converge in other metric spaces? The following examples and some of the exercises will illustrate.

Example 13.16: Let $\{x_n\}$ be a sequence of points in $\mathbb{R}^2$ equipped with the usual metric of Example 13.4. How can we recognize when the sequence $\{x_n\}$ converges to a point $z \in \mathbb{R}^2$? We recall that this means that

$$d_2(x_n, z) \to 0,$$

where d_2 is the usual metric in $\mathbb{R}^2$. It is instructive to see that this convergence is equivalent to coordinate-wise convergence.

If we write out the coordinates

$$x_n = (a_n, b_n), \quad z = (c, d),$$

then we can easily verify that this occurs precisely when the ordinary sequences $\{a_n\}$ and $\{b_n\}$ converge to c and d. Indeed just observe that

$$|a_n - c| \le d_2(x_n, z) \le |a_n - c| + |b_n - d|$$

and

$$|b_n - c| \le d_2(x_n, z) \le |a_n - c| + |b_n - d|$$

and this becomes obvious. (For a general version of this observation in the space $\mathbb{R}^n$, see Theorem 11.15.) ◀

Example 13.17: In the space $M[a,b]$ (Example 13.11), convergence reduces to what we called uniform convergence in Chapter 9. To see this, observe that for $f, g \in M[a,b]$

$$d(f,g) = \sup_{a\le t\le b} |f(t) - g(t)|.$$

Thus $d(f_k, g) \to 0$ if and only if

$$\sup_{a\le t\le b} |f_k(t) - g(t)| \to 0.$$

We saw (in Exercise 9.3.12) that this condition characterizes uniform convergence. ◀

Note. The definition of convergence implies that limit of the sequence must be a point in the space. Thus, for example, the sequence $\left\{\frac{1}{n}\right\}$ converges to 0 in $\mathbb{R}$, or in the subspace $[0,1]$, but does *not* converge in the subspace $(0,1)$ since 0 is not a point of (0,1). Similarly, if $x_n(t) = 1+t+\cdots+t^n$, then $x_n \to x_0$, where $x_0(t) = \frac{1}{1-t}$, in the space $M\left[0,\frac{1}{2}\right]$ or in the subspace $\mathcal{C}\left[0,\frac{1}{2}\right]$, but does *not* converge in the subspace $\mathcal{P}\left[0,\frac{1}{2}\right]$ of polynomials on $\left[0,\frac{1}{2}\right]$ because the function x_0 is not a polynomial.

Exercises

13.4.1 Describe the convergent sequences in a metric space (X,d), where d is the discrete metric.

SEE NOTE 295

13.4.2 Describe the convergent sequences in the euclidean n-dimensional space of Example 13.5.

SEE NOTE 296

13.4.3 Example 13.5 and Exercise 13.4.2 suggest that the following should be true for sequences in Hilbert space ℓ_2 (Example 13.7). In order for a sequence of points

$$x^{(n)} = (x_1^{(n)}, x_2^{(n)}, x_3^{(n)}, \dots)$$

$n = 1,2,3,\dots$ in ℓ_2 to converge to a point

$$y = \{y_1, y_2, y_3, \dots\}$$

in ℓ_2 it is necessary and sufficient that each $x_k^{(n)}$ converges to y_k as $n \to \infty$ for $k = 1,2,3,\dots$. Is this true?

SEE NOTE 297

13.4.4 **(The Hilbert Cube)** Consider the following subspace of Hilbert space ℓ_2:

$$H = \{(x_1, x_2, \dots) \in \ell_2 : |x_i| \le 1/i\}.$$

H is called the Hilbert cube. Show that, in contrast to Exercise 13.4.3, a sequence of points

$$x^{(n)} = (x_1^{(n)}, x_2^{(n)}, x_3^{(n)}, \dots)$$

in H converges if and only if each $x_k^{(n)}$ converges for $k = 1,2,3,\dots$.

13.4.5 Example 13.5 and Exercises 13.4.2 and 13.4.3 suggest that the following should be true for sequences in the function space $\mathcal{C}[0,1]$ consisting of the continuous functions on $[0,1]$ furnished with the metric

$$d(f,g) = \int_0^1 |f(t) - g(t)|\, dt.$$

It is a necessary (but perhaps not sufficient) condition for a sequence of functions $\{f_n\}$ to converge to a function g in $\mathcal{C}[0,1]$ that $f_n(x) \to g(x)$ for each $x \in [0,1]$. Is this true?

SEE NOTE 298

13.4.6 Establish some elementary properties of sequence convergence in a metric space:

(a) If $x_k \to x$ and $x_k \to y$, then $x = y$.

(b) If $x_k \to x$, then the sequence is bounded in the sense that

$$\sup\{d(x,x_k) : k = 1,2,3,4,\dots\} < \infty.$$

(c) Are there any further elementary results of the theory of real sequences that can be formulated and proved in a general metric space?

SEE NOTE 299

13.4.7 If $\{x_n\}$ and $\{y_n\}$ are convergent sequences in a metric space (X,d), show that $\lim_{n\to\infty} d(x_n,y_n)$ exists.

13.4.8 If $\{x_n\}$ and $\{y_n\}$ are sequences in a metric space (X,d) and

$$\lim_{n\to\infty} d(x_n,y_n) = 0,$$

show that they are either both convergent or both divergent.

13.4.9 Show that a sequence of real numbers $\{x_n\}$ is convergent in $\mathbb{R}$ if and only if it is convergent when $\mathbb{R}$ is furnished with any of the following metrics:

(a) $d(x,y) = |x-y|$ (i.e., the usual metric)

(b) $e_1(x,y) = \min\{1,|x-y|\}$

(c) $e_2(x,y) = \frac{|x-y|}{1+|x-y|}$

(Thus, while the usual metric and the metrics e_1 and e_2 differ, they describe the same class of convergent sequences.)

SEE NOTE 300

13.4.10 Generalize Exercise 13.4.9 to an arbitrary metric space.

SEE NOTE 301

13.4.11 Show that a sequence of points in the euclidean plane (Example 13.4) is convergent if and only if it is convergent under either of the metrics d_1 and d_∞ of Exercise 13.2.6. (Thus while the metrics d_1 and d_2 and d_∞ differ, they describe the same class of convergent sequences.)

13.4.12 Let a set X be equipped with two metrics d_1 and d_2. Determine the relation between sequence convergence in the two spaces (X,d_1) and (X,d_2) if one of the following conditions holds for some positive numbers m, M:

(a) $d_1(x,y) \le Md_2(x,y)$ for all $x, y \in X$.

(b) $md_1(x,y) \le d_2(x,y)$ for all $x, y \in X$.

(c) $md_1(x,y) \le d_2(x,y) \le Md_1(x,y)$ for all $x, y \in X$.

SEE NOTE 302

13.4.13 Consider the set $\mathcal{C}[0,1]$ of continuous functions on $[0,1]$ with the two metrics (both of which are important in analysis):

$$d_1(x,y) = \max_{0\le t\le 1} |x(t)-y(t)|$$

$$d_2(x,y) = \int_0^1 |x(t)-y(t)|\,dt$$

(a) If a sequence $\{x_k\}$ considered within the metric space $(\mathcal{C}[0,1],d_1)$ converges, must it also converge in the metric space $(\mathcal{C}[0,1],d_2)$? If it does converge in both metric spaces, must it necessarily converge to the same limit?

(b) What are the answers in (a) if we interchange the roles of d_1 and d_2?

13.4.14 Let $\mathcal{C}^1[a,b]$ denote the continuously differentiable functions on $[a,b]$. Define d by

$$d(x,y) = \max_{a\le t\le b} |x(t)-y(t)| + \max_{a\le t\le b} |x'(t)-y'(t)|$$

Verify that d is a metric on $\mathcal{C}^1[a,b]$. Which of the following sequences converge in the space $\mathcal{C}^1[0,1]$?

(a) $x_k = t^k$

(b) $x_k(t) = \dfrac{t}{k}$

(c) $x_k(t) = \dfrac{\sin(kt)}{k}$

(d) $x_k(t) = \displaystyle\int_0^{1/k} \sin(kst)\,ds$

SEE NOTE 303

13.4.15 In $\mathbb{R}$ we say that "addition is continuous," meaning that if $x_n \to x$ and $y_n \to y$, then $x_n + y_n \to x + y$. In each of the following spaces there is a natural way of defining addition. Define that addition and determine if addition is continuous in these spaces.

(a) ℓ_1

(b) ℓ_2

(c) ℓ_∞

(d) $\mathcal{C}[a,b]$

13.4.16 Let $\mathcal{K}$ denote the family of nonempty closed subsets of $[0,1]$ furnished with the Hausdorff metric of Exercise 13.3.9. Determine whether the following sequences converge and if so to what they converge.

(a) $A_n = [0, 1/n]$

(b) $B_n = \{1/n\}$

(c) $C_n = [1/2 - 1/n, 1/2 + 1/n]$ for $n \ge 2$

13.4.17 In the metric space $M[a,b]$ of Examples 13.11 and 13.17 we saw that convergence of sequences is precisely uniform convergence. Is it true then in the subspace $\mathcal{P}[a,b]$ of all polynomials on $[a,b]$ with the same metric that a sequence $\{p_n\}$ of polynomials converges if and only if it converges uniformly?

SEE NOTE 304

13.4.18 Using the theory of real sequences as a guide, formulate a definition for what should be meant by a "Cauchy sequence" in a metric space. Prove that every convergent sequence in a metric space must be Cauchy but that the converse need not be true.

SEE NOTE 305

13.5 Sets in a Metric Space

To develop the notion of convergence we need to extend to an arbitrary metric space some of the concepts we discussed for $\mathbb{R}$ in Chapter 3 and for $\mathbb{R}^n$ in Chapter 11. As a start, you may wish to draw pictures to illustrate these notions in $\mathbb{R}^2$. But to get a good sense of these notions we must have many examples. See the exercises at the end of this section.

1. For $x_0 \in X$ and $r > 0$, the set
$$B(x_0, r) = \{x \in X : d(x_0, x) < r\}$$
is called the *open ball* with center x_0 and radius r.

2. The set
$$B[x_0, r] = \{x \in X : d(x_0, x) \leq r\}$$
is called the *closed ball* with center x_0 and radius r.

3. A set $G \subset X$ is called *open* if for each $x_0 \in G$ there exists $r > 0$ such that $B(x_0, r) \subset G$.

4. A set $F \subset X$ is called *closed* if its complement $X \setminus F$ is open.

5. For a nonempty set E the *diameter* of E is the number
$$\sup\{d(x, y) : x, y \in E\},$$
which may be infinite.

6. A nonempty set E is *bounded* if
$$\sup\{d(x, y) : x, y \in E\}$$
is finite (i.e., if the set E has finite diameter).

7. A *neighborhood* of x_0 is any open set G containing x_0.

8. If $G = B(x_0, \varepsilon)$ we call G the ε-*neighborhood* of x_0.

9. A point x_0 is called an *interior point* of a set A if x_0 has a neighborhood contained in A.

10. The *interior* of a set A consists of all interior points of A and is denoted by A^o or, occasionally, $\text{int}(A)$.

11. A point $x_0 \in X$ is a *limit point* or *point of accumulation* of a set A if every neighborhood of x_0 contains infinitely many points of A.

12. An *isolated point* of a set A is a point x_0 that has a neighborhood G for which $G \cap A = \{x_0\}$.

13. The *closure* $\overline{A}$ of a set A consists of all points that are either in A or limit points of A.

14. A *boundary point* of A is a point x_0 such that every neighborhood of x_0 contains at least one point of A and at least one point of the complement $X \setminus A$.

15. Let A and B be subsets of X. If $\overline{A} \supset B$ or, equivalently, if every open ball centered at a point of B contains a point of A, we say that A is *dense* in B. (Note that this does not require A to be a subset of B.) If $\overline{A} = X$, we simply say that A is *dense*.

16. A set A in a metric space X is said to be *nowhere dense* (in X) if every open ball $B(x,\varepsilon)$ contains another open ball $B(y,\delta)$ such that $B(y,\delta) \cap A = \emptyset$.

17. The *distance* between a point $x \in X$ and a nonempty set $A \subset X$ is defined as
$$\text{dist}(x,A) = \inf\{d(x,y) : y \in A\}.$$

You will have noticed that the definitions we gave of concepts such as *open set, closed set, limit point* and the like are entirely analogous to the corresponding definitions we had already given in $\mathbb{R}$ and in $\mathbb{R}^n$. It should be no surprise that many of the elementary relations among these concepts that hold in $\mathbb{R}$ carry over to arbitrary metric spaces. We highlight some of the more important ones in the exercises that follow this section.

We consider a few examples.

Example 13.18: Let X be any nonempty set and let (X,d) be the discrete space (Example 13.3). Let $x_0 \in X$ be any point in the space. You should verify that each of the following statements is true.

1. $B(x_0,1) = \{x_0\}$
2. Every point is isolated in X
3. $B[x_0,1] = X$
4. Every set is both open and closed
5. X is bounded
6. Every set containing $x_0 \in X$ is a neighborhood of x_0
7. Every ε-neighborhood of x_0 is either $\{x_0\}$ or X
8. For every set $A \subset X$, $A^o = \overline{A} = A$, and A has no boundary points
9. X has no accumulation points
10. The only dense subset of X is X itself
11. The closure of an open ball $B(x,\varepsilon)$ is not necessarily the closed ball $B[x,\varepsilon]$

If any of these statements seem to contradict your intuition about metric spaces, be sure to restudy the definitions. ◀

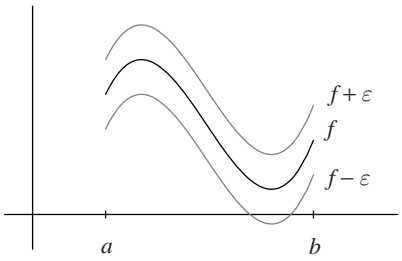


Figure 13.2. Ball centered at a function in $\mathcal{C}[a,b]$.

Example 13.19: Let K be the Cantor set viewed as a subspace of $[0,1]$. Let $\{(a_k,b_k)\}$ be the sequence of intervals complementary to K and let S consist of the midpoints of those intervals. For our metric space we take $X = K \cup S$ furnished with the usual real metric.

1. Every point of S is isolated, and no point of K is isolated.
2. K is closed.
3. S is open.
4. $\overline{S} = X$ so S is dense.
5. Each subset of S is open.

Again, be sure to check each of these statements. ◀

Example 13.20: Consider the space $\mathcal{C}[a,b]$ furnished with the supremum metric (see Example 13.10). Let $f \in \mathcal{C}[a,b]$ and let $\varepsilon > 0$.

1. $B(f,\varepsilon)$ consists of all continuous functions g that satisfy
$$|f(t) - g(t)| < \varepsilon$$
for all $t \in [a,b]$.
2. g is a boundary point of $B(f,\varepsilon)$ if and only if $|f(t) - g(t)| \leq \varepsilon$ for all $t \in [a,b]$ and there exists t_0 such that $|f(t_0) - g(t_0)| = \varepsilon$.
3. Geometrically, $g \in B(f,\varepsilon)$ if and only if the graph of g lies strictly between the graphs of $f - \varepsilon$ and $f + \varepsilon$.
4. Similarly, g is a boundary point of $B(f,\varepsilon)$ if and only if the graph of g lies between the graphs of $f - \varepsilon$ and $f + \varepsilon$ and there exists t_0 such that $g(t_0) = f(t_0) + \varepsilon$ or $g(t_0) = f(t_0) - \varepsilon$.

See Figure 13.2. ◀

Comparison Between Two Metrics Suppose X is a set furnished with two different metrics d_1 and d_2. We then have two metric spaces, (X,d_1) and (X,d_2). In general, we expect that the concepts introduced in this section will be different. For example, a set might be bounded or closed with respect to one metric but not the other. In practice, though, there are often close connections between the properties.

For example, it might happen that the class $\mathcal{G}_1$ of sets open with respect to d_1 is the same as the class $\mathcal{G}_2$ of sets open with respect to d_2. In this case, the two spaces (X,d_1) and (X,d_2) will have exactly the same class of convergent sequences, and the two spaces are the same in a sense we shall make precise in Section 13.6.2. In general, however, the two spaces are different: The set X is the same, but the classes of open sets are not, and sequences that converge in one of the spaces need not converge in the other. Exercises 13.4.13 and 13.5.13 provide important illustrations of this phenomenon.

Exercises

13.5.1 Show that the definition of a point of accumulation is equivalent to the following:

(a) x_0 is a point of accumulation of A if for every $\varepsilon > 0$ the set
$$A \cap B(x_0,\varepsilon) \setminus \{x_0\}$$
is nonempty.

(b) x_0 is a point of accumulation of A if there is a sequence of points $x_n \in A$ so that $x_n \neq x_0$ and $x_n \to x_0$.

(c) x_0 is a point of accumulation of A if $x \in \overline{A}$ and x_0 is not an isolated point of A.

(d) x_0 is a point of accumulation of A if x_0 is not an interior point of $(X \setminus A) \cup \{x_0\}$.

13.5.2 Show that every point of a set is either isolated or a point of accumulation.

13.5.3 Let $S = \{1/k : k = 1,2,3,\dots\}$ and furnish S with the usual real metric. Answer the following questions about this metric space.

(a) Which points are isolated in S?

(b) Which sets are open and which sets are closed in S?

(c) Which sets have a nonempty boundary?

(d) Which sets are dense in S?

(e) Which sets are nowhere dense in S?

13.5.4 Let E be a closed set in a metric space (X,d) and let x be a point that is not in E. Show that
$$\inf\{d(x,y) : y \in E\} > 0.$$
Show that if E and F are disjoint closed sets in a metric space, then it is not necessarily true that
$$\inf\{d(x,y) : x \in E,\ y \in F\} > 0.$$
SEE NOTE 306

13.5.5 Compute the diameter of the set
$$\{\{x_1,x_2,\dots\} \in \ell_2 : |x_i| \leq 1,\ i = 1,2,3,\dots\}$$
as a subset of the metric space ℓ_2.

13.5.6 Prove the following elementary property of sequence convergence in a metric space: It is true that $x_k \to x$ if and only if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $x_k \in B(x,\varepsilon)$ for all $k \geq N$.

13.5.7 A sequence in a metric space is *bounded* if the range of the sequence is a bounded set. On the real line $\mathbb{R}$ every bounded sequence has a convergent subsequence. Is a similar statement true in any metric space?

SEE NOTE 307

13.5.8 Show, in a general metric space, that the open ball is open and that the closed ball is closed. Give an example of a metric space in which a closed ball $B[x,\varepsilon]$ is not necessarily the closure of the open ball $B(x,\varepsilon)$.

13.5.9 We chose to define a closed set as one whose complement is open. Show that the following are equivalent for a subset A of a metric space (X,d):

(a) $X \setminus A$ is open.

(b) A contains all its limit points.

(c) $A = \overline{A}$.

13.5.10 Let (X,d) be a metric space.

(a) Prove that X and $\emptyset$ are both open and closed.

(b) Prove that a finite union of closed sets is closed and a finite intersection of open sets is open.

(c) Prove that an arbitrary union of open sets is open and an arbitrary intersection of closed sets is closed.

13.5.11 Let X denote the set of points

$$\{0\} \cup \{1/k : k = 1,2,3,\dots\}$$

in $\mathbb{R}$ furnished with the usual real metric. Answer the following questions:

(a) Which points are isolated in X?

(b) Which sets are open and which sets are closed?

(c) Which sets are both open and closed?

(d) Is X bounded?

(e) Which sets have a nonempty boundary?

(f) Does X have any accumulation points?

(g) Describe all dense subsets of X.

(h) Is the closure of an open ball $B(x,\varepsilon)$ necessarily the closed ball $B[x,\varepsilon]$?

13.5.12 Which of the following subsets of $M[a,b]$ are closed?

(a) $\mathcal{C}[a,b]$

(b) $\mathcal{P}[a,b]$, the polynomials on $[a,b]$

(c) $\mathcal{R}[a,b]$, the integrable functions on $[a,b]$

(d) $\mathcal{P}_n[a,b]$, the polynomials of degree $\leq n$ on $[a,b]$

(e) $\triangle[a,b]$, the differentiable functions on $[a,b]$

(f) $\triangle'[a,b]$, the derivatives of differentiable functions with bounded derivatives on $[a,b]$

SEE NOTE 308

13.5.13 Consider the set $\mathcal{C}[0,1]$ of continuous functions on $[0,1]$ with the two metrics:

$$d_1(x,y) = \max_{0\le t\le 1} |x(t)-y(t)|$$
$$d_2(x,y) = \int_0^1 |x(t)-y(y)|\,dt.$$

Let B_1 be an open ball in $(\mathcal{C}[0,1],d_1)$ and let B_2 be an open ball in $(\mathcal{C}[0,1],d_2)$.

(a) Is B_1 open in $(\mathcal{C}[0,1],d_2)$?

(b) Is B_2 open in $(\mathcal{C}[0,1],d_1)$?

13.5.14 Which of these are closed subsets of the metric space ℓ_∞ of bounded sequences (Example 13.9)?

(a) ℓ_2

(b) ℓ_1

(c) c, the set of all convergent sequences of real numbers

(d) c_0, the set of all those sequences that converge to 0

(e) p, the set of all bounded sequences of nonnegative real numbers

(f) r, the set of all bounded sequences of rational numbers

13.5.15 **(The Hilbert Cube)** Consider the following two subsets of Hilbert space ℓ_2:

(a) $H = \{(x_1,x_2,\dots) \in \ell_2 : |x_i| \le 1/i\}$

(b) $G = \{(x_1,x_2,\dots) \in \ell_2 : |x_i| < 1/i\}$

H is called the Hilbert cube. Show that H is closed in ℓ_2. Is G open?

SEE NOTE 309

13.5.16 Which of the properties of closure and interior found in Exercises 4.3.6 and 4.3.7 are valid in every metric space? In particular, give an example of a metric space in which the examples sought in part (d) of these exercises don't exist.

13.5.17 Let $X = \mathbb{R}^2$. Sketch the unit spheres, that is, the set $\{x : d(x,0) = 1\}$, for each of the following metrics, each defined for all $x = (x_1,x_2)$ and $y = (y_1,y_2)$ in $\mathbb{R}^2$.

$$d_1(x,y) = |x_1-y_1| + |x_2-y_2|$$
$$d_2(x,y) = \sqrt{(x_1-y_1)^2+(x_2-y_2)^2}$$
$$d_\infty(x,y) = \max\{|x_1-y_1|,|x_2-y_2|\}$$

13.5.18 Let (X_1,d_1) and (X_2,d_2) be metric spaces and let $X_1 \times X_2$ be the product space as defined in Exercise 13.2.9.

(a) Show that every set of the form $A \times B$, where A is an open subset of (X_1,d_1) and B is an open subset of (X_2,d_2) is an open subset of $X_1 \times X_2$.

(b) Show that every open subset of $X_1 \times X_2$ can be expressed as a union

$$\bigcup_{i \in I} A_i \times B_i,$$

where each A_i is an open subset of (X_1, d_1) and B_i is an open subset of (X_2, d_2).

13.5.19 State and prove a version of the Bolzano-Weierstrass theorem that is valid in $\mathbb{R}^2$.
SEE NOTE 310

13.5.20 ✂ State and prove a version of the Heine-Borel theorem that is valid in $\mathbb{R}^2$.
SEE NOTE 311

13.5.21 A set A in a metric space X is called

(a) *nowhere dense* (in X) if every open ball $B(x, \varepsilon)$ contains another open ball $B(y, \delta)$ such that $B(y, \delta) \cap A = \emptyset$

(b) *dense in itself* if every point of A is a limit point of A

(c) *perfect* if it is closed and dense in itself

The following are several subsets A of certain metric spaces (X, d). Which are nowhere dense? Which are dense in themselves? Which are perfect.

(a) $A = (0, 1)$ and $X = \mathbb{R}$

(b) $A = (0, 1)$ and $X = (0, 1)$

(c) $A = \mathcal{C}[a, b]$ and $X = M[a, b]$

(d) $A =$ constant functions and $X = \mathcal{C}[a, b]$

(e) $A = \{x \in c_0 : x_k = 0$ for all but finitely many values of $k\}$ and $X = c_0$ (see Exercise 13.5.14)

(f) $A = \{(x_1, x_2) \in \mathbb{R}^2 : x_1^2 + x_2^2 = 1\}$ and $X = \mathbb{R}^2$

(g) A is the collection of nonempty closed subsets of [0,1] with no more than k elements and $X = \mathcal{K}$ is the family of nonempty closed subsets of [0,1] furnished with the Hausdorff metric (Exercise 13.3.9)

(h) A is the union of the families of all sets in part (g) taken over $k = 1, 2, 3, \ldots$ and $X = \mathcal{K}$ is the same as in part (g)

13.6 Functions

Our study of limits and continuity for functions $f : \mathbb{R} \to \mathbb{R}$ can be generalized by studying these same notions for functions mapping a metric space into another. Let us begin with some examples of functions from one metric space to another.

Example 13.21: We define a function $f : \mathbb{R}^2 \to \mathbb{R}$ by

$$f(x_1, x_2) = \frac{x_1 x_2}{x_1^2 + x_2^2} \quad f(0, 0) = 0.$$

We would naturally be interested in the properties such a function would have when $\mathbb{R}^2$ and $\mathbb{R}$ are furnished with the usual euclidean metrics (see Example 13.5). Thus we consider this a function with domain one metric space and range in another metric space. ◀

Example 13.22: We can interpret the operation of integration as a function mapping the metric space $\mathcal{C}[a,b]$ into $\mathbb{R}$ by

$$T(f) = \int_a^b f(t)\,dt.$$

Thus integration can be considered a real-valued function on a metric space of functions. ◀

Example 13.23: We define a function S mapping the metric space $\mathcal{C}[a,b]$ into itself by

$$(S(f))(t) = \int_a^t f(s)\,ds.$$

Thus the notion of an indefinite integral is realized as an operation or function on the metric space $\mathcal{C}[a,b]$ with values in $\mathcal{C}[a,b]$. ◀

Example 13.24: The operation of differentiation can be interpreted as a function on an appropriate metric space. Let $\mathcal{C}^1[0,1]$ consist of those functions on [0,1] with continuous derivatives. We define a function $D : \mathcal{C}^1[0,1] \to \mathcal{C}[0,1]$ by

$$D(f) = f'.$$

If we use the sup metrics on these spaces, then we can interpret the operation of differentiation as a function from one metric space into another. ◀

Example 13.25: If f is a continuous function on the interval $[0,\pi]$, we write its Fourier sine series as

$$\sum_{k=1}^{\infty} b_k \sin kx,$$

where the coefficients are given by Fourier's formulas

$$b_k = \frac{2}{\pi}\int_0^{\pi} f(t)\sin kt\,dt.$$

With a slight shift in viewpoint this operation can be considered in the context of metric spaces. The input function f is transformed into a sequence of numbers $\{b_k\}$. Thus we can write it as $F(f) = \{b_k\}$ on the understanding that the terms of the sequence $\{b_k\}$ are given by the formulas.

But what sequence space is appropriate? One of the elementary inequalities of Fourier series shows which space to use. It is not difficult to prove that

$$\sum_{k=1}^{\infty} b_k^2 \le \frac{2}{\pi}\int_0^{\pi} f^2(t)\,dt.$$

(This is called Bessel's inequality.) Evidently, we can interpret F as a mapping from $\mathcal{C}[0,\pi]$ into the sequence space ℓ_2. ◀

In each of these five examples we have introduced a function $f : X \to Y$ from one metric space (X,d) to another (Y,e). In Examples 13.22, 13.23, and 13.24 we follow a common practice of using uppercase letters (T, S and D) when the metric spaces are also vector spaces. We often use the terms *transformation* or *operator* in place of the term *function*.

Our study of functions between metric spaces begins with continuity. We then proceed to study two special kinds of continuous functions between metric spaces, isometries and homeomorphisms. Homeomorphisms are functions that are not merely continuous but whose inverses are continuous. Isometries are continuous functions that preserve distances.

13.6.1 Continuity

We wish to know, as in the case of functions from $\mathbb{R}$ to $\mathbb{R}$, whether these examples of functions (transformations) defined previously are "continuous." In defining continuity of functions between metric spaces we try to capture the following idea:

> If $T : X \to Y$, then T is continuous at $x_0 \in X$ provided that all points near x_0 are mapped to points near $T(x_0)$.

We make this precise in exactly the same way as we did for functions from subsets of $\mathbb{R}$ to $\mathbb{R}$ (in Section 5.1.2) and for functions from subsets of $\mathbb{R}^n$ to $\mathbb{R}^m$ (in Section 11.7).

Definition 13.26: Let (X,d) and (Y,e) be metric spaces and let $T : X \to Y$. We say that T is *continuous* at $x_0 \in X$ if for every sequence $\{x_k\}$ converging to $x_0 \in X$ the sequence $\{T(x_k)\}$ converges to $T(x_0) \in Y$. If T is continuous at every point of X, we say that T is *continuous*.

You can verify that the alternate characterizations of continuity that were valid for real functions, namely the ε-δ characterization and the neighborhood characterization, are also valid here. The proofs are the same. We list these characterizations for reference.

Theorem 13.27: *Let (X,d) and (Y,e) be metric spaces and let $T : X \to Y$.*

(i) T is continuous at x_0 if and only if for every neighborhood V of $T(x_0)$ there exists a neighborhood U of x_0 such that $T(U) \subset V$.

(ii) T is continuous at x_0 if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $e(T(x),T(x_0)) < \varepsilon$ whenever $d(x,x_0) < \delta$.

An important special case is the following. For continuity everywhere it is the way in which open sets are treated that is most significant. Note that it is not the image set $T(G)$ that is to be open when $G \subset X$ is open, but the preimage $T^{-1}(G)$ when $G \subset Y$ is open.

Theorem 13.28: *Let (X,d) and (Y,e) be metric spaces and let $T : X \to Y$. Then T is continuous at each point of X if and only if for each open set $G \subset Y$, the set*

$$T^{-1}(G) = \{x : T(x) \in G\}$$

is open in X.

You may wish to draw pictures in a familiar setting (such as $X = Y = \mathbb{R}^2$) to illustrate these characterizations.

Example 13.29: Let us look again at Example 13.21. Here

$$f(x_1,x_2) = \frac{x_1x_2}{x_1^2+x_2^2} \quad f(0,0) = 0.$$

Let us check continuity at $(0,0)$. Since $f(0,0) = 0$ (by definition), f will be continuous at (0,0) if and only if

$$\lim_{n\to\infty} f(u_n, v_n) = 0$$

for every sequence of points $(u_n, v_n) \to (0,0)$. Any one example of a sequence for which this fails shows that f is not continuous there. For example, observe that $f(1/n, 1/n) \to 1/2$. [Perhaps we should redefine $f(0,0)$ to be $1/2$; in that case then again f is not continuous since $f(1/n,0) \to 0$. Thus no value of $f(0,0)$ can make this function continuous there.]

At every other point it is easy to check that f is continuous. Take any sequence of points $(u_n, v_n) \to (a,b) \neq (0,0)$ and verify by the usual elementary sequence methods that $f(u_n, v_n) \to f(a,b)$. ◀

Example 13.30: Let us look again at Example 13.22. Here

$$T(f) = \int_a^b f(t)\,dt.$$

If $f_k \to f$ in X, then $f_k \to f$ uniformly on $[a,b]$. By Theorem 9.26

$$\int_a^b f_k(t)\,dt \to \int_a^b f(t)\,dt,$$

that is, $T(f_k) \to T(f)$ in $\mathbb{R}$. Thus T is continuous at f. Since this is true for all $f \in \mathcal{C}[a,b]$, T is continuous on $\mathcal{C}[a,b]$. ◀

Example 13.31: Let us look again at Example 13.23. We verify that S defined by

$$(S(f))(t) = \int_a^t f(s)\,ds$$

is continuous at every $f \in \mathcal{C}[a,b]$. Let $f_k \to f$ in that metric space. This means that

$$d(f_k, f) = \max_t |f_k(t) - f(t)| \to 0 \text{ as } k \to \infty$$

(that is, $f_k \to f$ uniformly on $[a,b]$). You should verify each step in these calculations:

$$\begin{aligned} d(S(f_k), S(f)) &= \max_t |(S(f_k))(t) - (S(f))(t)| \\ &= \max_t \left| \int_a^t (f_k(s) - f(s))\,ds \right| \\ &\leq \max_t \int_a^t |f_k(s) - f(s)|\,ds = \int_a^b |f_k(s) - f(s)|\,ds \\ &\leq (b-a) \max_t |f_k(t) - f(t)| = (b-a)d(f_k, f). \end{aligned}$$

Since $\lim_{k\to\infty} d(f_k, f) = 0$ by hypothesis, we conclude that

$$\lim_{k\to\infty} d(S(f_k), S(f)) = 0.$$

Thus $S(f_k) \to S(f)$, and S is continuous. ◀

Example 13.32: Let us look again at Example 13.24. We ask if the function $D : \mathcal{C}^1[0,1] \to \mathcal{C}[0,1]$ defined by $D(f) = f'$ is continuous. We must recall the metrics used. We use the sup metric on both spaces $\mathcal{C}^1[0,1]$ and $\mathcal{C}[0,1]$. Thus $f_k \to f$ if and only if $f_k \to f$ uniformly on $[0,1]$. A specific example suffices to show that D is not continuous. For each $k \in \mathbb{N}$, let $f_k(t) = t^k/k$. Then $f_k \to 0$ in $\mathcal{C}^1[0,1]$. In order for D to be continuous we must have $D(f_k) \to D(0) = 0$ in $\mathcal{C}[0,1]$. But

$$D(f_k)(t) = f_k'(t) = t^{k-1}$$

and this sequence does not converge in $\mathcal{C}[0,1]$. (Had we imposed a different metric on $\mathcal{C}^1[0,1]$ the answer might have been different.) ◀

Observe that Examples 13.22 and 13.23 involve operators defined by integrals. Such operators are often (but not always) continuous. Example 13.24 shows a differential operator. For such operators, continuity often fails. (But see Exercise 13.6.16 at the end of this section.)

Exercises

13.6.1 Let (X,d) be a discrete space.

(a) What functions $f : X \to \mathbb{R}$ are continuous everywhere?

(b) What functions $f : \mathbb{R} \to X$ are continuous everywhere?

SEE NOTE 312

13.6.2 Verify statements (i) and (ii) of Theorem 13.27 that provide characterizations of continuity.

13.6.3 Prove Theorem 13.28.

13.6.4 Prove that $T : X \to Y$ is continuous if and only if $T^{-1}(E)$ is closed for every closed set $E \subset Y$.

13.6.5 In Section 5.4.4 we defined continuity of a function $f : A \to \mathbb{R}$ when A is a subset of $\mathbb{R}$. Verify that the definition given there agrees with the definition given in this section when A is considered as a metric space with $d(x,y) = |x-y|$.

13.6.6 Verify each inequality and each equality in the calculation found in Example 13.31.

13.6.7 Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by writing

$$f(x_1,x_2) = \frac{x_1^2 x_2}{x_1^4 + x_2^2} \quad f(0,0) = 0.$$

Show that $\lim_{x\to 0} f(x,mx) = 0$ for every $m \in \mathbb{R}$, but f is discontinuous at $(0,0)$.

SEE NOTE 313

13.6.8 Let $f : \mathbb{R}^2 \to \mathbb{R}$. Suppose f is *separately continuous*, that is, $f(x_1, v)$ is a continuous function of v for every $x_1 \in \mathbb{R}$ and $f(u, x_2)$ is a continuous function of u for every $x_2 \in \mathbb{R}$. If the continuity with respect to one of the variables is uniform with respect to the other variable, then f is continuous. Make this statement precise and prove it.
SEE NOTE 314

13.6.9 Let (X, d) be a metric space. Prove that d is continuous on $X \times X$, where $X \times X$ is furnished with the product metric. See Exercise 13.2.9.

13.6.10 Let (X_1, d_1) and (X_2, d_2) be metric spaces and let $X_1 \times X_2$ be the product space as defined in Exercise 13.2.9. Show that the functions $f : X_1 \times X_2 \to X_1$ and $g : X_1 \times X_2 \to X_2$ defined by $f(x, y) = x$ and $g(x, y) = y$ are continuous.

13.6.11 Let (X, d) be a metric space and let A be a nonempty subset of X. Define $f : X \to \mathbb{R}$ by

$$f(x) = \operatorname{dist}(x, A) = \inf\{d(x, y) : y \in A\}.$$

(a) Show that $|f(x) - f(y)| \leq d(x, y)$ for all $x, y \in X$.

(b) Show that f defines a continuous real-valued function on X.

(c) Show that $\{x \in X : f(x) = 0\} = \overline{A}$.

(d) Show that $\{x \in X : f(x) > 0\} = \operatorname{int}(X \setminus A)$.

(e) Show that, unless X contains only a single point, there exists a continuous real-valued function defined on X that is not constant.

(f) If $E \subset X$ is closed and $x_0 \notin E$, show that there is a continuous real-valued function g on X so that $g(x_0) = 1$ and $g(x) = 0$ for all $x \in E$.

(g) If E and F are disjoint closed subsets of X, show that there is a continuous real-valued function g on X so that $g(x) = 1$ for all $x \in F$ and $g(x) = 0$ for all $x \in E$.

(h) If E and F are disjoint closed subsets of X, show that there are disjoint open sets G_1 and G_2 so that $E \subset G_1$ and $F \subset G_2$.

(i) In the special case where X is the real line with the usual metric and K denotes the Cantor ternary set, sketch the graph of the function $f(x) = \operatorname{dist}(x, K)$.

(j) Give an example of a metric space, a point x_0, and a set $A \subset X$ so that $\operatorname{dist}(x_0, A) = 1$ but so that $d(x, x_0) \neq 1$ for every $x \in \overline{A}$.

SEE NOTE 315

13.6.12 Let x and y be real-valued functions on $[0, 1]$. Define $f : [0, 1] \to \mathbb{R}^2$ by $f(t) = (x(t), y(t))$. If f is continuous, then f is called a *continuous curve*. Prove that f is a continuous curve if and only if both functions x and y are continuous.
SEE NOTE 316

13.6.13 Prove that the class of continuous real-valued functions on a metric space is closed under the arithmetic operations of addition, subtraction, and multiplication. (How about division?)

13.6.14 State precisely and prove a theorem that asserts under which conditions the composition $f \circ g$ of two continuous functions is continuous.

13.6.15 Prove that the function of Example 13.25 is continuous.
SEE NOTE 317

13.6.16 Let $\mathcal{C}^1[a,b]$ consist of the continuously differentiable functions on $[a,b]$. Define for $f, g \in \mathcal{C}^1[a,b]$

$$d(f,g) = \max_{a\le t\le b} |f(t)-g(t)| + \max_{a\le t\le b} |f'(t)-g'(t)|.$$

(a) Prove that d is a metric.

(b) Let $D : \mathcal{C}^1[a,b] \to \mathcal{C}[a,b]$ be defined by $D(f) = f'$. Prove that D is continuous. (Here, as usual, $\mathcal{C}[a,b]$ has the sup metric.)

13.6.17 Let $\mathcal{C}[0,1]$ consist of the continuous functions on $[0,1]$ and furnished with the metric

$$d(f,g) = \int_0^1 |f(t)-g(t)|\,dt.$$

Define $T : \mathcal{C}[0,1] \to \mathbb{R}$ by

$$T(f) = \int_0^1 f(t)\,dt.$$

Is T continuous?

13.6.18 Extend the following concepts and results from functions of one variable to functions of two variables:

(a) Define *uniform continuity* for a function $f : \mathbb{R}^2 \to \mathbb{R}$.

(b) Prove that a continuous real-valued function f defined on a closed and bounded subset $K \subset \mathbb{R}^2$ is uniformly continuous on K.

(c) Prove that a continuous real-valued function f defined on a closed and bounded subset $K \subset \mathbb{R}^2$ is bounded on K.

(d) Prove that a continuous real-valued function f defined on a closed and bounded subset $K \subset \mathbb{R}^2$ achieves an absolute maximum on K.

SEE NOTE 318

13.6.19 Let $\{f_k\}$ be a sequence of real-valued functions on a metric space X. Define what it means for $\{f_k\}$ to approach a function f uniformly. Show that if each of the functions f_k is continuous and $f_k \to f$ uniformly on X, then f is continuous on X.

13.6.2 Homeomorphisms

A *homeomorphism* between metric spaces is a one-to-one onto mapping that is continuous and whose inverse is also continuous.

To motivate our discussion of homeomorphisms let us consider two problems that have been suggested in several of the exercises. First, suppose that a set X is furnished with two different metrics d_1 and d_2. While the spaces (X,d_1) and (X,d_2) may be different as metric spaces, it is still possible that they have closely related properties. When could we recognize that they have the same open sets, the same closed sets, the same convergent sequences, etc.? The following example discusses this problem in a concrete setting.

Example 13.33: Consider the three metrics d_1, d_2, and d_∞ on the plane $\mathbb{R}^2$ defined by

$$\begin{aligned} d_1(x,y) &= |x_1-y_1| + |x_2-y_2|, \\ d_2(x,y) &= \sqrt{(x_1-y_1)^2+(x_2-y_2)^2}, \\ d_\infty(x,y) &= \max\{|x_1-y_1|, |x_2-y_2|\}, \end{aligned}$$

for points $x = (x_1, x_2)$, $y = (y_1, y_2)$. (We have seen these metrics in Example 13.4 and Exercise 13.2.6.)

While these are three different metrics on $\mathbb{R}^2$ we will see that the open sets are the same under the three metrics and that convergent sequences are also identical. Why is this so?

An examination of the unit balls centered at the origin helps explain why. (Our analysis here repeats some of the discussion in Section 11.6.2 which you may have skipped.) Figure 11.4 on page 456 shows the three unit balls centered at the origin for these metrics. More generally, an open ball centered at x with radius r with respect to the metric d_∞ will be the inside of a square with sides parallel to the coordinate axes. Its center will be at x and its side length will be $2r$. For the metric d_2 the ball will be the inside of a circle of radius r, and for d_1 the ball will be the inside of a square of side length $r\sqrt{2}$ with sides parallel to the lines $x_2 = \pm x_1$.

Let us denote the balls in the three spaces

$$(\mathbb{R}^2, d_1),\ (\mathbb{R}^2, d_2),\ \text{ and }\ (\mathbb{R}^2, d_\infty)$$

by $B_1(x,r)$, $B_2(x,r)$ and $B_\infty(x,r)$, respectively. It is easy to verify analytically that for every $x \in \mathbb{R}^2$ and $r > 0$

$$B_1(x,r) \subset B_2(x,r) \subset B_\infty(x,r). \tag{5}$$

Furthermore, for every $x \in \mathbb{R}^2$ and $r > 0$ there exists $s > 0$ such that

$$B_\infty(x,s) \subset B_1(x,r) \tag{6}$$

(Exercise 13.6.24). It follows that any ball centered at x with respect to one of the three metrics contains a ball centered at x with respect to either of the other two metrics. Thus any set that is open with respect to one of the metrics is also open with respect to the other two. ◀

This fact in our example, that the three different metrics give rise to the same family of open sets, has important consequences. If the open sets are the same, then the convergent sequences are the same. If the open sets are the same, then the continuous functions are the same. Thus any sequence $\{x_n\}$ converging to a point x_0 with respect to one of the metrics also converges to x_0 with respect to the other two. Further, a function f mapping $\mathbb{R}^2$ into a metric space (Y, e) will be continuous with respect to one of the metrics if and only if it is continuous with respect to to the others.

We could summarize our example by stating that the three metric spaces $(\mathbb{R}^2, d_1)$, $(\mathbb{R}^2, d_2)$, and $(\mathbb{R}^2, d_\infty)$ have the same "topological properties:" From the topological point of view they are indistinguishable. We need to make this notion precise. The key observation in our example concerned the open sets relative to each of the three metrics. The *topology* of a metric space is simply the family of open sets. Thus, loosely, a topological property is one that can be expressed in terms of the open sets and need not be expressed directly in terms of the metric defined on the space.

Before we state our definitions, consider the second problem. Suppose that two metric spaces (X, d) and (Y, e) are closely related in the sense that there is a one-to-one onto function $h : X \to Y$. Thus $Y = h(X)$ is a "copy" of X and each point $x \in X$ relates to a unique point $y = h(x) \in Y$. The problem is this: What additional properties should

h have so that there is the same relation between the open sets of X and the open sets of Y, namely that G is an open subset of X precisely if $h(G)$ is an open subset of Y? The answer to this is given in the definition.

Definition 13.34: Let (X,d) and (Y,e) be metric spaces. A function

$$h : X \to Y$$

is called a *homeomorphism* if h meets the following conditions:

1. h is one-to-one,
2. h maps X onto Y,
3. h is continuous, and
4. h^{-1} is continuous.

Condition (4) is equivalent to h being an *open map*, that is, h maps open sets in X onto open sets in Y (Theorem 13.28). Two metric spaces are said to be *homeomorphic* or *topologically equivalent* if there is a homeomorphism mapping one space onto the other. A property that is preserved under homeomorphisms is called a *topological property*.

Some topological properties are listed in Exercise 13.6.21. Here are some examples of spaces that are topologically equivalent.

Example 13.35: The spaces $X = (-1,1)$ and $Y = \mathbb{R}$, both furnished with the usual metrics, are topologically equivalent. [For an appropriate homeomorphism take $h(x) = \tan \pi x/2$.] ◀

Example 13.36: The space

$$X = \{(x_1,x_2) \in \mathbb{R}^2 : x_2 = 1/x_1,\ x_1 > 0\}$$

furnished with the euclidean metric and the interval $Y = (0,\infty)$ are topologically equivalent. [For a homeomorphism take $h(x_1,x_2) = x_1$.] ◀

Example 13.37: The three spaces $(\mathbb{R}^2,d_1)$, $(\mathbb{R}^2,d_2)$, and $(\mathbb{R}^2,d_\infty)$ of Example 13.33 are topologically equivalent. Between any pair take h as the identity map. Assertions 5 and 6 can be used to prove that h is a homeomorphism. Because the identity map is a homeomorphism we can say more: The open sets are the same for each of the three metrics. ◀

Note. In the last example the three metrics give rise to exactly the same families of open sets of $\mathbb{R}^2$. It is not always true, however, that if d and e are two metrics on a set X, and (X,d) is homeomorphic to (X,e), then d and e induce the same family of open sets. (See Exercise 13.6.30.) What *is* true is that if $h : (X,d) \to (X,e)$ is a homeomorphism, then a set $A \subset X$ is open in (X,d) if and only if the set $h(A)$ is open with respect to (X,e).

Sometimes it is not immediately clear whether two spaces are homeomorphic. One way to show that the spaces are *not* homeomorphic is to exhibit a topological property possessed by one of the spaces but not by the other.

Example 13.38: For instance, an open interval (a,b) in $\mathbb{R}$ cannot be homeomorphic to a closed interval $[c,d]$ (with respect to the usual metrics). The property that is not preserved is the Bolzano-Weierstrass property: Any sequence in $[c,d]$ must contain a convergent subsequence, but this is not true in (a,b). ◀

Example 13.39: $\mathbb{Q}$ cannot be homeomorphic to $\mathbb{R} \setminus \mathbb{Q}$. The property that is not preserved is countability: A homeomorphism cannot map a countable set to an uncountable one. ◀

We present some further examples of homeomorphic spaces in the exercises.

Exercises

13.6.20 Show that topologically equivalence is an equivalence relation, that is, prove the following:

(a) X is homeomorphic to itself.

(b) If X is homeomorphic to Y, then Y is homeomorphic to X.

(c) If X is homeomorphic to Y and Y is homeomorphic to Z, then X is homeomorphic to Z.

13.6.21 The following are several properties a set A in a metric space X might possess. Verify that each of these is a topological property; that is, if A has the property in (X,d), and (Y,e) is homeomorphic to (X,d) via the homeomorphism h, then $h(A)$ has the same property in (Y,e).

(a) A is open

(b) A is closed

(c) A is dense

(d) A is nowhere dense

(e) A is countable

13.6.22 The following are several properties a set A in a metric space X might possess. Verify that each of these is a *not* a topological property. (Thus while these properties are defined in terms of the metric, they are not invariant under homeomorphisms.)

(a) A is bounded

(b) A has diameter equal to 1

(c) For any pair $x, y \in A$ there is an element $z \in A$ with $d(x,z) = d(y,z)$

(d) All points in A are equidistant from some point $z \in X$, that is, $d(x,z) = d(y,z)$ for all $x, y \in A$

13.6.23 In Example 13.35, $h^{-1}(\mathbb{R}) = (-1,1)$ which is an open interval. Does this violate our claim that h^{-1} maps closed sets onto closed sets?

13.6.24 Verify the two assertions (5) and (6) of Example 13.33.

13.6.25 Show that if $h : X \to Y$ is a homeomorphism, then a sequence $\{x_k\}$ converges in X if and only if the sequence $\{h(x_k)\}$ converges in Y.

13.6.26 Show that $\mathbb{R}$ is homeomorphic to $(0,1)$ but not to $[0,1]$.
SEE NOTE 319

13.6.27 Show that $\mathbb{R}$ is homeomorphic to $(0,\infty)$ but not to $[0,\infty)$.
SEE NOTE 320

13.6.28 Let (X,d) be a metric space. Show that there is a metric e on X so that (X,d) and (X,e) are topologically equivalent and such that X has a finite diameter in (X,e).
SEE NOTE 321

13.6.29 Let (X,e) be a metric space and suppose that

$$\inf\{e(x,y) : x,y \in X,\ x \neq y\} > 0.$$

Show that (X,e) is topologically equivalent to (X,d), where d is the discrete metric on X. Is the converse true?
SEE NOTE 322

13.6.30 Let $X = (0,1) \cup (2,3)$. Define metrics d and e on X as follows. If $x_1 \neq x_2$, then

$$d(x_1,x_2) = \begin{cases} |x_1 - x_2|, & \text{if } x_1,x_2 \in (0,1) \\ 1, & \text{otherwise} \end{cases}$$

and

$$e(x_1,x_2) = \begin{cases} |x_1 - x_2|, & \text{if } x_1,x_2 \in (2,3) \\ 1, & \text{otherwise.} \end{cases}$$

Find a homeomorphism $h : (X,d) \to (X,e)$. Show the identity is not a homeomorphism between the spaces. Exhibit a set that is open with respect to d but not with respect to e and a sequence $\{x_k\}$ that converges with respect to d but not with respect to e.

13.6.31 A careless student states, "If d and e are metrics on a nonempty set X and the metric spaces (X,d) and (X,e) are homeomorphic, the two spaces have exactly the same open sets and the identity map is a homeomorphism between them. Indeed let $h : X \to X$ be a homeomorphism. Then h^{-1} is also a homeomorphism between the two spaces, so $h^{-1} \circ h$ is also a homeomorphism. But $h^{-1} \circ h$ is just the identity. It follows, too, that all the open sets are the same." In view of Exercise 13.6.30 this cannot be true. Where is the flaw in the argument?

13.6.32
(a) Let $X = (0,1)$, $Y = (2,3) \cup (4,5)$, both with the usual metrics. Prove that X and Y are not homeomorphic.

(b) Show that if X and Y are furnished with the discrete metric, they *are* homeomorphic.

13.6.33 Let $f : [a,b] \to \mathbb{R}$ be continuous. Show that the interval $[a,b]$ is homeomorphic to the graph of the function f, that is, to the set

$$\{(x,y) : x \in [a,b],\ y = f(x)\}$$

considered as a subset of $\mathbb{R}^2$.

13.6.34 For $x = (x_1,x_2)$, $y = (y_1,y_2)$ in $\mathbb{R}^2$ and $1 < p < \infty$ define

$$d_p(x,y) = (|x_1 - y_1|^p + |x_2 - y_2|^p)^{1/p}.$$

It can be shown that d_p is a metric on $\mathbb{R}^2$.

(a) Sketch the unit balls $B_p(0,1)$ centered at the origin for several values of p. (We already did this for $p=1$ and $p=2$ in Fig. 11.4.) Observe that $B_p(0,1)$ seems to approach $B_\infty(0,1)$.

(b) Prove analytically that for all $x,y \in \mathbb{R}^2$,

$$\lim_{p\to\infty} d_p(x,y) = d_\infty(x,y).$$

(c) Show that all the spaces $(\mathbb{R}^2, d_p)$ are topologically equivalent.

13.6.35 Show that the open interval $(0,1)$ is homeomorphic with the set

$$\left\{(x_1,x_2) \in \mathbb{R}^2 : x_2 = \sin\frac{1}{x_1},\ x_1 > 0\right\}$$

when $\mathbb{R}$ and $\mathbb{R}^2$ are furnished with the euclidean metrics.

13.6.36 Let (X,e) be a metric space. Show that (X,e) is topologically equivalent to (X,d), where d is the discrete metric if and only if any one of the following properties holds:

(a) Any intersection of a family of open sets is open.

(b) For any open set G the closure $\overline{G}$ is also open.

(c) Every point x in X is isolated.

13.6.37 A metric space X is called *connected* if it cannot be expressed as a disjoint union of two nonempty open sets.

(a) Prove that if $f : X \to Y$ is continuous and X is connected, then so is Y.

(b) Show that connectedness is a topological property,

(c) Characterize the connected subspaces of $\mathbb{R}$.

SEE NOTE 323

13.6.38 Let (X,d) be a connected metric space (see Exercise 13.6.37). Show that X either contains only one point or else uncountably many points.

SEE NOTE 324

13.6.39 Show that a metric space (X,d) is connected if and only if every continuous function $f : X \to \mathbb{R}$ has the intermediate value property.

13.6.40 Two points a and b in a metric space (X,d) can be *connected by a curve* if there is a continuous function $f : [0,1] \to X$ so that $f(0) = a$ and $f(1) = b$. Show that if every pair of points in X can be connected by a curve, then X is connected. Is the converse true?

SEE NOTE 325

13.6.41 Show $\mathbb{R}$ and $\mathbb{R}^2$ are not homeomorphic.

SEE NOTE 326

13.6.42 Show that the Cantor set K in $\mathbb{R}$ is homeomorphic to $K \times K$ in $\mathbb{R}^2$.

SEE NOTE 327

13.6.43 **(Cantor Space)** Denote by $2^{\mathbb{N}}$ the set of all sequences $x_0, x_1, x_2, \ldots$ of 0's and 1's furnished with the metric

$$d(x,y) = \sum_{k=0}^{\infty} \frac{|x_k - y_k|}{2^k}.$$

(a) Verify that $(2^{\mathbb{N}}, d)$ is a metric space.

(b) Show that if $x_k = y_k$ for all $k = 0, 1, 2, \ldots, n$, then

$$d(x, y) \leq 1/2^n$$

and if

$$d(x, y) \leq 1/2^n,$$

then $x_k = y_k$ for all $k \leq n$.

(c) Show that $(2^{\mathbb{N}}, d)$ is homeomorphic to the Cantor set.

SEE NOTE 328

13.6.44 Let X and Y be closed and bounded subsets of $\mathbb{R}$ each having exactly one limit point.

(a) Prove that X and Y are homeomorphic.

(b) Part (a) establishes the fact that there exists a homeomorphism

$$h : X \to Y.$$

Is it necessarily true that there exists a homeomorphism $h : \mathbb{R} \to \mathbb{R}$ such that $h(X) = Y$?

(c) Is it necessarily true that there exists a homeomorphism $h : \mathbb{R}^2 \to \mathbb{R}^2$ that maps $X \times \{0\}$ to $Y \times \{0\}$?

SEE NOTE 329

13.6.45 Show that there is a homeomorphism h of the Cantor set such that $h(0) = 1/3$ and $h(1/3) = 0$. Does there exist a homeomorphism h of $\mathbb{R}$ onto $\mathbb{R}$ such that h maps the Cantor set onto itself for which $h(0) = 1/3$ and $h(1/3) = 0$?

13.6.3 Isometries

When two metric spaces (X, d) and (Y, e) are topologically equivalent they share certain properties—if one space has a property, so does the other. Topological properties of sets in X carry over to their images in Y. Examples of topological properties were given in Exercises 13.6.21 and 13.6.37. We shall study in Sections 13.7 and 13.12 two further topological properties, separability and compactness, that are important concepts in the theory of metric spaces.

But two topologically equivalent spaces can still have strikingly different properties. [For example, while $(0, 1)$ and $(0, \infty)$ are topologically equivalent, one is bounded and the other not.] What stronger notion of equivalence captures the idea that the spaces have identical metric space properties?

This stronger form of equivalence of metric spaces involves the concept of *isometry* or *congruence*. Recall that in elementary geometry we learn, for example, if the three sides of a triangle T_1 in a plane have the same lengths as the sides of a triangle T_2, then T_1 and T_2 are congruent. This means that T_1 can be *rigidly moved* onto the triangle T_2. We can make this notion of rigid motion precise in the general setting of metric spaces.

Definition 13.40: Let (X, d) and (Y, e) be metric spaces. A function h mapping X onto Y is called an *isometry* if

$$e(h(x), h(y)) = d(x, y)$$

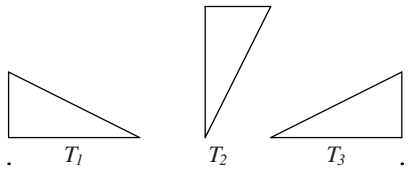


Figure 13.3. The triangles T_1, T_2 and T_3 are isometric.

for all $x, y \in X$. The two spaces are called *isometric* if there is an isometry of one onto the other. A property is called a *metric property* if it is preserved by isometries.

Observe that an isometry is just a homeomorphism that preserves distance. Thus two congruent triangles in the plane are isometric metric spaces when they are viewed as subspaces of the plane.

Note. We have to be a bit careful in our use of the term *rigid motion*. Consider the right triangles in Figure 13.3. Triangles T_1 and T_2 are congruent—the isometry h such that $h(T_1) = T_2$ can be a translation t followed by a rotation r: $h = r \circ t$. Thus T_1 is moved onto T_2 while staying in the plane. But there is no such motion (i.e., combination of translations and rotations) within $\mathbb{R}^2$ that maps T_1 onto T_3. Nonetheless, T_1 and T_3 are isometric. You may wish to show how, for example, we can construct an isometry between T_1 and T_3. (An isometry of $\mathbb{R}^3$ onto $\mathbb{R}^3$ can do the job, but this isn't necessary. All we need is an isometric mapping of T_1 onto T_3. The domain of that mapping need be only T_1, not some larger space.) Compare these remarks with Exercises 13.6.44 and 13.6.45.

Let's look at a few examples.

Example 13.41: Any two open intervals in $\mathbb{R}$ are homeomorphic. They are isometric if and only if they have the same length. ◀

Example 13.42: The two subspaces $\mathbb{N}$ and $\mathbb{N} \cup \{0\}$ of $\mathbb{R}$ are isometric. [Use the function $h(x) = x - 1$.] ◀

Example 13.43: Here is a more interesting example of spaces that are isometric. Recall the three metrics d_1, d_2, and d_∞ on $\mathbb{R}^2$ that we discussed at the beginning of Section 13.6.2. The spaces $(\mathbb{R}^2, d_1)$ and $(\mathbb{R}^2, d_\infty)$ are isometric. This might appear a bit surprising for the identity map is *not* an isometry. Instead take the function

$$h(x_1, x_2) = \left(\frac{x_1 + x_2}{2}, \frac{x_1 - x_2}{2} \right)$$

for $(x_1, x_2) \in \mathbb{R}^2$. We leave verification that h is an isometry as Exercise 13.6.50. ◀

Example 13.44: But $(\mathbb{R}^2, d_2)$ is *not* isometric to $(\mathbb{R}^2, d_\infty)$. To see this, observe that the d_∞ distance between any pair of the four points $\left(\pm\frac{1}{2}, \pm\frac{1}{2}\right)$ is 1 and the images of these points under an isometry would have the same property with respect to d_2. This is impossible. (See Exercise 13.6.49.) ◀

Example 13.45: $\mathbb{R}$ and $\{(x_1,x_2) \in \mathbb{R}^2 : x_2 = 0\}$ are isometric. [Use the function $h(x) = (x,0)$.] ◀

Embeddings Example 13.45 asserts that the real line and the x-axis in $\mathbb{R}^2$ are isometric spaces. Thus inside the space $\mathbb{R}^2$ is an isometric copy of the real line. We cannot say that the space $\mathbb{R}^2$ *contains* the real line, just that it contains an identical copy of the real line. In order to have some language with which to discuss such situations we introduce the notion of an embedding.

An *embedding* of a metric space X into another space Y is an isometric copy of the space X that is situated in Y. Thus we have $h(X) \subset Y$ with X and $h(X)$ isometric. Even though the two metric spaces are distinct and may contain formally no elements in common we can use the embedding notion to imagine that one is, for all practical purposes, a subspace of the other.

Exercises

13.6.46 Show that isometry is an equivalence relation, that is, prove the following:

(a) X is isometric to itself.

(b) If X is isometric to Y, then Y is isometric to X.

(c) If X is isometric to Y and Y is isometric to Z, then X is isometric to Z.

13.6.47 Is the following statement true? If X is isometric to a subspace of Y and Y is isometric to a subspace of X, then X and Y are isometric.
SEE NOTE 330

13.6.48 Prove that every isometry is also a homeomorphism but not conversely.

13.6.49 Show that a three-point discrete space can be embedded in $\mathbb{R}^2$ but that a four-point discrete space cannot.

13.6.50 Verify that the function

$$h(x_1,x_2) = \left(\frac{x_1+x_2}{2}, \frac{x_1-x_2}{2}\right)$$

is an isometry of $(\mathbb{R}^2, d_\infty)$ and $(\mathbb{R}^2, d_1)$. The Jacobian of this mapping is not 1. Does this violate anything you learned about Jacobians?

13.6.51 Show that $\mathbb{R}$ and $\mathbb{R}^2$ are not isometric (when they have the usual metrics). Find metrics e_1 and e_2 on $\mathbb{R}$ and $\mathbb{R}^2$ such that $(\mathbb{R}, e_1)$ is isometric to $(\mathbb{R}^2, e_2)$.

13.6.52 Let $X = (0,\infty)$ and $Y = \{(x_1,x_2) \in \mathbb{R}^2 : x_2 = 1/x_1, x_1 > 0\}$. Are X and Y isometric (using the usual metrics for $\mathbb{R}$ and $\mathbb{R}^2$)?
SEE NOTE 331

13.6.53 Let f be an increasing function on the interval $[0,1]$. Let $d(x,y) = |x-y|$ be the ordinary metric on $X = [0,1]$ and let e be the metric defined by

$$e(x,y) = |f(x) - f(y)|.$$

(a) Under what conditions on f are the spaces (X,d) and (X,e) topologically equivalent?

(b) Under what conditions on f are the spaces (X,d) and (X,e) isometric?

(c) Is there any choice of f so that (X,e) is topologically equivalent to X furnished with the discrete metric?

13.7 Separable Spaces

In our studies of elementary analysis in the earlier chapters we occasionally made use of the fact that the rational numbers formed a dense subset of $\mathbb{R}$. This was convenient for two reasons: First, the rational numbers are much easier to describe and handle than the real numbers, and second, they formed a countable subset. Generally, in metric space theory the existence of a countable dense subset within a metric space makes many arguments much simpler. This leads us to the following terminology.

Definition 13.46: A metric space (X,d) is said to be *separable* if it possesses a countable dense subset.

A separable space is "not too large" in the sense that some countable set can be used to approximate all members of the space. To show a space is separable we must show that there exists a countable dense subset.

Example 13.47: The space $\mathbb{R}$ of real numbers is separable. For a countable dense set take the rationals $\mathbb{Q}$. (There are many other countable dense sets in $\mathbb{R}$.) ◀

Example 13.48: The space $\mathbb{R}\setminus\mathbb{Q}$ of irrational numbers is separable. For a countable dense set take, for example, the set of all numbers of the form $m\sqrt{2}/n$, where m and n are integers. (Note that we cannot take $\mathbb{Q}$ this time since it is not a *subset* of $\mathbb{R}\setminus\mathbb{Q}$.) ◀

Example 13.49: The space $\mathbb{R}^n$ is separable. Take $\mathbb{Q}^n$ as a countable dense subset. ◀

Example 13.50: Let X be a nonempty countable set with any metric d. Then (X,d) is separable. For a countable dense set take X itself. ◀

Example 13.51: A famous theorem of Weierstrass states that every function that is continuous on the interval $[a,b]$ can be approximated uniformly by polynomials. (See Section 10.8.4.) Thus, given such a function f and an $\varepsilon > 0$ there is a polynomial p so that $|f(x)-p(x)| < \varepsilon$ for all $x \in [a,b]$. But any such polynomial p can itself be approximated by a polynomial with rational coefficients q merely by adjusting each of the coefficients, so that $|p(x)-q(x)| < \varepsilon$ for all $x \in [a,b]$. Putting these together we have $|f(x)-q(x)| < 2\varepsilon$ for all $x \in [a,b]$. Thus the class $\mathcal{P}_q$ of polynomials with rational coefficients is dense in $\mathcal{C}[a,b]$. Here, as usual, $\mathcal{C}[a,b]$ is furnished with the metric

$$d(f,g) = \max_{a\le t\le b} |f(t)-g(t)|.$$

Observe that $\mathcal{P}_q$ is countable, so $\mathcal{C}[a,b]$ possesses a countable dense subset and hence is a separable metric space. ◀

Example 13.52: The space c of convergent sequences of real numbers is a separable subspace of the space ℓ_∞ (which is itself not separable, as we shall argue later). To see that c is separable, let A_n denote the set of all sequences of the form

$$(r_1, r_2, r_3, \ldots, r_{n-1}, r_n, r_n, r_n, \ldots),$$

where each r_1, r_2, …, r_n is a rational number. Let $A = \bigcup_{n=1}^{\infty} A_n$. You can verify (Exercise 13.7.2) that A is a countable dense subset of c. ◀

Some Nonseparable Examples We can often show that a space is *not* separable by exhibiting an uncountable disjoint family of open sets. Since a dense set must intersect every nonempty open set, there could not exist a countable dense set.

Example 13.53: Any uncountable set X furnished with the discrete metric is not separable since each singleton set is open. No countable set can be dense. ◄

Example 13.54: Consider the space $M[a,b]$ of bounded functions on $[a,b]$ furnished with the sup metric (see Example 13.11). This space is larger than the separable subspace $C[a,b]$. To see that this space is not separable, observe that if f and g are characteristic functions of distinct sets, then $d(f,g) = 1$. There are uncountably many distinct subsets of $[a,b]$ so the space $M[a,b]$ contains uncountably many disjoint open balls and is therefore nonseparable. ◄

Example 13.55: The space ℓ_∞ of bounded sequences is not separable. To see this, let

$$A = \{\{x_i\} : x_i = 0 \text{ or } x_i = 1\}.$$

This set is an uncountable subset of ℓ_∞. If $x, y \in A$ with $x \neq y$, then $d(x,y) = 1$. Thus the family

$$\{B(x, 1/2) : x \in A\}$$

is an uncountable disjoint family of open balls in ℓ_∞. Any dense set in ℓ_∞ must intersect each ball of this family and so must be uncountable. ◄

Separability Is a Topological Property Any space that is topologically equivalent to a separable metric space must be itself separable. We can obtain this from a slightly more general statement about how dense sets are preserved under continuous mappings.

Theorem 13.56: *Let (X,d) and (Y,e) be metric spaces and let $f : X \to Y$ be a continuous function. If D is dense in X, then the set $f(D)$ is dense in $f(X)$.*

Proof. Let V be a nonempty open set in $f(X)$. Since f is continuous, the set $U = f^{-1}(V)$ is open in X. Since D is dense in X, there exists $x \in D \cap U$. Thus $f(x) \in V$. It follows that $f(D)$ is dense in $f(X)$. ■

Corollary 13.57: *If (X,d) is a separable metric space and (Y,e) is homeomorphic to (X,d), then (Y,e) is also separable. Thus, separability is a topological property of metric spaces.*

Exercises

13.7.1 Show that a metric space (X,d) is separable if and only if for every $\varepsilon > 0$ there is a countable set C_ε so that every point in the space is closer than ε to some point in C_ε.

13.7.2 Show that the set A in Example 13.52 is countable and dense in c.

13.7.3 Show that the space $\mathcal{K}$ of closed nonempty subsets of $[0,1]$ with the Hausdorff metric (Exercise 13.3.9) is separable.

SEE NOTE 332

13.7.4 Show that the set of polygonal functions on $[a,b]$ with rational vertices is a countable dense subset of $\mathcal{C}[a,b]$.
SEE NOTE 333

13.7.5 Prove that a subspace of a separable space is separable.
SEE NOTE 334

13.7.6 Let c_0 denote the subspace of ℓ_∞ consisting of those sequences that converge to 0.

(a) Use Exercise 13.7.5 and Example 13.52 to show that c_0 is separable.

(b) Show that c_0 is separable by exhibiting a countable dense subset of c_0.

13.7.7 Let (X,d) be the space of Example 13.14. Show that this space is separable by showing any dense subset of $\mathcal{C}[a,b]$ is also dense in (X,d).

13.7.8 By a unit sphere we mean a set

$$\{x : d(x,x_0) = 1\}.$$

(a) Show that the unit sphere $\{x : d(x,0) = 1\}$ is separable in $\mathbb{R}^2$ with respect to any of the metrics d_1, d_2, d_∞.

(b) Give an example of a metric space and a unit sphere that is not separable.

SEE NOTE 335

13.7.9 Prove that a metric space X is separable if and only if there exists a countable collection $\mathcal{U}$ of open sets such that every open set in X can be expressed as a union of members of $\mathcal{U}$.

13.7.10 Prove that a product of two separable metric spaces, furnished with the product metric, is separable.

13.7.11 Let $X = \mathbb{R}$ and let d be the discrete metric on X. Determine which of the metric spaces ℓ_∞, $\mathcal{C}[0,1]$, or $M[0,1]$ contains an isometric copy of (X,d).
SEE NOTE 336

13.8 Complete Spaces

In our study of sequences in Chapter 2 we introduced the concept of a Cauchy sequence of real numbers and showed that every such sequence converges to some real number. In our study of uniform convergence in Chapter 9 we saw that a similar result is valid for a uniformly Cauchy sequence of functions (Theorem 9.12). This notion of a *Cauchy sequence* can be defined in any metric space.

Definition 13.58: Let (X,d) be a metric space and let $\{x_k\}$ be a sequence in X. This sequence is called a *Cauchy sequence* if for each $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that if $m \geq N$ and $n \geq N$, then $d(x_n,x_m) < \varepsilon$.

The statement in the definition is equivalent to the requirement that

$$\lim_{m,n\to\infty} d(x_m,x_n) = 0.$$

Example 13.59: In the metric space $M[a,b]$ a sequence $\{f_k\}$ is a Cauchy sequence if and only if the sequence of functions $\{f_k\}$ is uniformly Cauchy according to Definition 9.11. Thus Theorem 9.12 can be interpreted as stating that every Cauchy sequence in $M[a,b]$ converges in $M[a,b]$. ◀

It is not the case in a general metric space that all Cauchy sequences converge. This is so on the real line, and Example 13.59 shows that it is true in the space $M[a,b]$. A metric space that does have this property is said to be complete. We have already used this word to describe a certain property of the real numbers. In fact, completeness of $\mathbb{R}$, which we interpreted in Chapter 1 by the least upper bound property, is exactly equivalent to the fact that Cauchy sequences converge.

Definition 13.60: A metric space (X,d) is *complete* if every Cauchy sequence in X converges (to an element of X).

We know that $\mathbb{R}$ and $M[a,b]$ are complete. Let's look at a few familiar examples of spaces that are not complete and observe why this is so.

Example 13.61: The space $\mathbb{Q}$, with the usual metric of $\mathbb{R}$, is not complete. Every Cauchy sequence in $\mathbb{Q}$ does converge to some real number, but not necessarily to a member of $\mathbb{Q}$. For example, the sequence $\{(1+1/n)^n\}$ is a Cauchy sequence of rational numbers and yet does not converge in $\mathbb{Q}$. ◀

Example 13.62: The space $\mathcal{P}[0,1]$ of polynomials on $[0,1]$, furnished with the sup metric is not complete. For example, the sequence $\{p_k\}$ with

$$p_k(t) = 1 + t + \frac{t^2}{2} + \cdots + \frac{t^k}{k!}$$

converges uniformly to the function e^t on [0,1]. This follows from our study of uniform convergence of power series in Section 10.3. Thus $\{p_k\}$ is a Cauchy sequence in the space $M[0,1]$ (Exercise 13.8.1). It follows that it is a Cauchy sequence in the subspace $\mathcal{P}$ of $M[0,1]$. But $\{p_k\}$ does not converge in $\mathcal{P}$, since the function e^t is not a polynomial. ◀

13.8.1 Completeness Proofs

How can we show that a given metric space (X,d) is complete? Sometimes certain theorems (such as Theorem 13.64, presented later in this section) can be applied to give a proof. But often we must simply use the definition and show directly that every Cauchy sequence in X converges in X. This can be achieved by applying the following steps to an arbitrary Cauchy sequence $\{x_k\}$ in X.

1. Find a natural "candidate" x_0 for the limit of the sequence.
2. Show that this candidate is in the space X.
3. Verify that $x_k \to x_0$.

It is important to realize that step (2) is essential. For example, as we observed in Example 13.61, the sequence $\{(1+1/n)^n\}$ is a Cauchy sequence in the space $\mathbb{Q}$ with a natural candidate for a limit, namely the number e, but e is *not* in $\mathbb{Q}$.

Example 13.63: To show how this process works, let us give a direct proof that $M[a,b]$ is complete using exactly these steps. (In Example 13.59 we obtained this as a consequence of Theorem 9.12; now we can obtain Theorem 9.12 instead as a consequence of the completeness of $M[a,b]$.)

Recall that the metric here is the sup metric

$$d(f,g) = \sup_{a\le t\le b} |f(t)-g(t)|,$$

and convergence reduces to what we called uniform convergence in Chapter 9. Let $\{f_k\}$ be a Cauchy sequence in $M[a,b]$.

Step 1. We wish to find a natural candidate for the limit. A bit of reflection on the meaning of uniform convergence suggests that we consider limits of the form $\lim_{k\to\infty} f_k(t)$ for each $t\in[a,b]$. We observe that for every $t\in[a,b]$ the sequence $\{f_k(t)\}$ is a Cauchy sequence of real numbers. This follows immediately from the inequality

$$|f_n(t)-f_m(t)| \le \sup_{a\le s\le b} |f_n(s)-f_m(s)| = d(f_n,f_m).$$

Since $\mathbb{R}$ is complete,

$$f_0(t) = \lim_{k\to\infty} f_k(t)$$

exists for each $t\in[a,b]$. This limit defines a function f_0 on $[a,b]$. The function f_0 is our candidate.

Step 2. We must verify that $f_0\in M[a,b]$. To do this, we must show that f_0 is a bounded function. Observe that the fact that f_0 is by its definition the pointwise limit of a sequence of bounded functions does not in itself guarantee that f_0 is bounded. It is the fact that $\{f_k\}$ is a Cauchy sequence that guarantees that f_0 is bounded.

To see this, choose $N\in\mathbb{N}$ such that $d(f_m,f_n)<1$ for all $m,n\ge N$. Then

$$|f_N(t)-f_m(t)|\le 1$$

for all $m\ge N$ and $t\in[a,b]$. Letting $m\to\infty$ in this inequality, we see that

$$|f_N(t)-f_0(t)|\le 1$$

for all $t\in[a,b]$. Since $f_N\in M[a,b]$, f_N is bounded, say $|f_N(t)|\le A$ for all $t\in[a,b]$. Then $|f_0(t)|\le A+1$ for all $t\in[a,b]$, so f_0 is bounded, and therefore a member of $M[a,b]$.

Step 3. We must show that $f_k\to f_0$ in the space $M[a,b]$; that is, $f_k\to f_0$ uniformly on $[a,b]$. Again, it is not enough to know that $f_k\to f_0$ pointwise. We used pointwise convergence to get a candidate, knowing that if there is to be a limit, it must be the pointwise limit. Thus we still need to verify the convergence is uniform.

Let $\varepsilon>0$. Since $\{f_k\}$ is a Cauchy sequence in $M[a,b]$, there exists $N\in\mathbb{N}$ such that if $n\ge N$, then

$$d(f_n,f_N)\le \frac{\varepsilon}{2},$$

that is,

$$|f_N(t) - f_n(t)| < \frac{\varepsilon}{2} \text{ for all } t \in [a,b]. \tag{7}$$

Thus, for all $t \in [a,b]$,

$$|f_N(t) - f_0(t)| = \lim_{m\to\infty} |f_N(t) - f_m(t)| \leq \frac{\varepsilon}{2}. \tag{8}$$

It follows from (7) and (8) that, for $n \geq N$ and for $t \in [a,b]$,

$$|f_n(t) - f_0(t)| \leq |f_n(t) - f_N(t)| + |f_N(t) - f_0(t)| < \varepsilon.$$

This proves that $f_k \to f_0$ uniformly on $[a,b]$. ◀

13.8.2 Subspaces of a Complete Space

Suppose now we wish to prove that the space $\mathcal{C}[a,b]$ is complete. We could argue exactly as we did with $M[a,b]$, but it is easier to consider that $\mathcal{C}[a,b]$ is a subspace of the complete space $M[a,b]$. What property should a subspace have so that it too is complete? Theorem 13.64 supplies the answer and is a useful tool in checking for completeness of many spaces that we might encounter.

Theorem 13.64: *Let X be a complete metric space and let Y be a subspace of X. Then Y is complete if and only if Y is closed in X.*

Proof. Suppose first that Y is closed and $\{y_k\}$ is a Cauchy sequence in Y. Since X is complete, $\{y_k\}$ converges to some point $x_0 \in X$. But Y is closed, so $x_0 \in Y$. Thus Y is complete.

Conversely, suppose that Y is complete. We show Y is closed. Let x_0 be a limit point of Y. Then there exists a sequence $\{y_k\}$ in Y such that $y_k \to x_0$. This sequence, being convergent in X, is a Cauchy sequence in X, and therefore in Y. Since Y is complete, the sequence $\{y_k\}$ converges to a point y_0 in Y. But limits are unique, so $y_0 = x_0$. Thus $x_0 \in Y$ and Y is closed. ■

13.8.3 Cantor Intersection Property

✂ *Enrichment section. May be omitted.*

Here is another criterion for completeness. You will recognize that this theorem is a direct generalization of the familiar Cantor intersection theorem in $\mathbb{R}$ (Section 4.5.2).

Theorem 13.65: *A metric space (X,d) is complete if and only if the intersection of every descending sequence of closed balls whose radii approach zero consists of a single point.*

We leave the proof as Exercise 13.8.8. See also Exercise 13.8.14 which shows that the requirement on the radii cannot be dropped. Thus, when the radii of the balls gets small, the intersection consists of a single point. But if the radii remain large, the intersection, instead of being large, might be empty!

You should also check the following useful version:

A metric space (X,d) is complete if and only if the intersection of every descending sequence of closed sets whose diameters approach zero consists of a single point.

13.8.4 Completion of a Metric Space

✂ *Enrichment section. May be omitted.*

In all of our examples of metric spaces that were not complete we have simply chosen a subset of a complete metric space that was not closed. In that way our subset would have Cauchy sequences that do not converge in the subset. Examples 13.61 and13.62 were like this. It can be shown that this is, in a sense, the *only* way that a Cauchy sequence can fail to converge. It is always the case that an incomplete space resides within a larger complete metric space and Cauchy sequences in the former space that do not converge must, however, converge to a point in the larger space.

More precisely, if (X,d) is a metric space, there is always a complete metric space in which X can be isometrically *embedded*. Thus there exists a complete metric space (Y,e) and a function $h : X \to Y$ such that h maps X isometrically onto $h(X)$. Furthermore, this can be achieved in such a way that $(h(X),e)$ is dense in (Y,e). The space (Y,e) is called the *completion* of (X,d) and is *unique up to isometry*. This means that if (Y',e') is any other complete metric space into which (X,d) can be embedded as a dense subspace, then (Y',e') and (Y,e) are isometric

Let us state this formally as a theorem that we shall not prove.[1].

Theorem 13.66: *Let (X,d) be a metric space. Then there exists a complete metric space (Y,e) and a function $h : X \to Y$ such that h maps X isometrically onto $h(X)$, which has the property that $(h(X),e)$ is dense in (Y,e). The space (Y,e) with this property is unique up to an isometry and is called the* completion *of (X,d).*

We can clearly consider the space $\mathbb{R}$ as the completion of the space $\mathbb{Q}$ of rational numbers. Note, however, that the goal of embedding $\mathbb{Q}$ as a dense subset of a larger complete metric space would be more ambitious than just this theorem might indicate: The algebraic and order structures would also need to be preserved, as indeed they are for $\mathbb{R}$.

Example 13.67: Since the space $\mathcal{P}[a,b]$ has been shown in Example 13.62 to be incomplete, what space might be used for its completion? A theorem of Weierstrass[2] implies that the completion of $\mathcal{P}[a,b]$ is $\mathcal{C}[a,b]$. ◀

Exercises

13.8.1 Show that a convergent sequence in a metric space must be a Cauchy sequence.
SEE NOTE 337

[1] Exercise 13.15.16 can be made a basis for a proof. Another method, using Cauchy sequences, can be found in Bruckner, Bruckner, and Thomson, *Real Analysis,* Prentice Hall (1997), §9.6.7.

[2] See the material in Section 10.8.4; for a different proof see Bruckner, Bruckner, and Thomson, op. cit., §9.13.

13.8.2 Show that every Cauchy sequence in a metric space is bounded.
SEE NOTE 338

13.8.3 Is the following statement true? A sequence $\{x_n\}$ in a metric space (X,d) is Cauchy if and only if $\lim_{n\to\infty} d(x_n,x_{n+1}) = 0$.
SEE NOTE 339

13.8.4 Let $\{x_n\}$ and $\{y_n\}$ be Cauchy sequences in a metric space (X,d). Show that $d(x_n,y_n)$ converges even if the sequences $\{x_n\}$ and $\{y_n\}$ themselves do not.
SEE NOTE 340

13.8.5 Let $\{x_n\}$ be a Cauchy sequence in a metric space (X,d). Show that there must be a subsequence $\{x_{n_k}\}$ with these two properties:

(a) $d(x_{n_{k+1}},x_{n_k}) < 2^{-k}$ for $k = 0,1,2,\ldots$.

(b) $B[x_{n_1},1] \supset B[x_{n_2},1/2] \supset \cdots \supset B[x_{n_k},1/2^{k-1}] \supset \ldots$.

SEE NOTE 341

13.8.6 Prove that if any subsequence of a Cauchy sequence in a metric space converges, then the full sequence also converges.
SEE NOTE 342

13.8.7 Show that a metric space (X,d) is complete if and only if every sequence of points $\{x_n\}$ in X with the property that

$$\sum_{k=1}^{\infty} d(x_k,x_{k+1}) < \infty$$

converges.
SEE NOTE 343

13.8.8 Prove Theorem 13.65.
SEE NOTE 344

13.8.9 Let X be a nonempty set and let d be the discrete metric. Is (X,d) complete?

13.8.10 Obtain that each of the following spaces is complete by applying Theorem 13.64:

(a) $\mathcal{C}[a,b]$

(b) The Cantor set

(c) $\mathbb{N}$

13.8.11 Obtain that each of the following spaces is not complete by applying Theorem 13.64:

(a) The space $\mathcal{P}[0,1]$ of polynomials on $[0,1]$

(b) The set $\{1,1/2,1/3,1/4,\ldots\}$ with the usual real metric

(c) The open interval $(0,1)$

13.8.12 Show that euclidean n-dimensional space $\mathbb{R}^n$ is complete. (See Example 13.5.)

13.8.13 Show that the space $\mathcal{K}$ of Exercise 13.3.9 is complete.
SEE NOTE 345

13.8.14 **(Sierpinski's space)** Let $\mathbb{N}$ be furnished with the metric

$$d(m,n) = \begin{cases} \frac{1}{m+n} + 1, & \text{if } m \neq n \\ 0, & \text{otherwise.} \end{cases}$$

(a) Verify that d is a metric on $\mathbb{N}$.

(b) Show that every subset of $(\mathbb{N}, d)$ is open.

(c) Show that $(\mathbb{N}, d)$ is complete.

(d) Let $S_n = \left\{ m \in \mathbb{N} : d(m,n) \leq 1 + \dfrac{1}{2n} \right\}$.
Show that $S_n = \{n, n+1, n+2, \dots\}$.

(e) Show that $\{S_n\}$ is a descending sequence of closed balls whose intersection is empty.

(f) Reconcile part (e) with Theorem 13.65.

13.8.15 Verify that the spaces c (Exercise 13.5.14) and ℓ_∞ (Example 13.9) are complete. Is the subspace c_0 of c complete?

13.8.16 Let (X,d) and (Y,e) be metric spaces and let f be a continuous function mapping X onto Y.

(a) If X is separable, must Y be separable?

(b) If X is complete, must Y be complete?

(c) Is separability a topological property? Is completeness?

(d) Do the answers to (a) and (b) change if f is an isometry?

SEE NOTE 346

13.8.17 Let (X_1, d_1) and (X_2, d_2) be complete metric spaces. Is the product space $X_1 \times X_2$ also complete? (See Exercise 13.2.9 for the definition of the product metric.)

13.8.18 A metric space X is said to be *absolutely closed* if every isometric image of X into a space Y is closed in Y. Show that X is absolutely closed if and only if it is complete.

SEE NOTE 347

13.9 Contraction Maps

Let f be a continuous function mapping an interval $[a,b]$ into itself. Then f has a *fixed point*; that is, there exists a point $x \in [a,b]$ such that $f(x) = x$. To see this, observe that if a and b are not fixed points, then $f(a) > a$ and $f(b) < b$. The function $g(t) = f(t) - t$ satisfies $g(a) > 0 > g(b)$. Since g is continuous, the intermediate value theorem applies to show that there exists a point x such that $g(x) = 0$ (Theorem 5.53). From this it follows that $f(x) = x$.

A comparable statement is not valid for continuous functions mapping $\mathbb{R} \to \mathbb{R}$.

Example 13.68: The function $f(x) = x + 1$ is a function mapping $\mathbb{R}$ into itself and has no fixed points since $f(x) = x$ for no value of x. ◀

The existence of fixed points of mappings has proved to be of considerable importance in analysis. We could ask just what it takes about a space X that every continuous function $f : X \to X$ should have a fixed point. For example, a famous theorem of Luitzen Brouwer (1881–1966) asserts that any closed sphere in $\mathbb{R}^n$ would have this property. Instead we restrict our attention by considering not all continuous functions, but ones that are contractions in a sense that will be defined. For example, if a function $f : \mathbb{R} \to \mathbb{R}$ satisfies the Lipschitz condition

$$|f(x) - f(y)| \leq \alpha|x - y|$$

for all $x, y \in \mathbb{R}$, with $0 < \alpha < 1$, then f is "contractive" in the sense that any two points x and y move to points $f(x)$ and $f(y)$ that are closer together. This condition guarantees that f will have a fixed point.

Example 13.69: The functions $f(x) = x/2$ or $g(x) = \sin(x/2)$ map $\mathbb{R} \to \mathbb{R}$ and both satisfy such a Lipschitz condition with $\alpha = 1/2$. (Just check the derivatives and apply the mean value theorem.) In this case the fixed points of f and g are easy to find: Look for the point of intersection of the graph of f (or g) with the line $y = x$. ◀

This statement about the existence of fixed points for certain Lipschitz functions is a special case of an important property of complete metric spaces. The property, often called Banach's fixed point theorem, states that every *contraction map* of a complete metric space into itself has a unique fixed point. Let us give precise definitions for these terms.

Definition 13.70: Let (X, d) be a metric space and let $A : X \to X$. If there exists a number $\alpha \in (0, 1)$ such that

$$d(A(x), A(y)) \leq \alpha d(x, y) \text{ for all } x, y \in X,$$

we say A is a *contraction map*.

Definition 13.71: Let (X, d) be a metric space and let $A : X \to X$. A point $x \in X$ for which $A(x) = x$ is called a *fixed point*.

There are a few immediate properties of contraction maps that we should establish. The first shows that if a contraction map has a fixed point, then it has only one.

Lemma 13.72: *Let A be a contraction map that has a fixed point. Then that fixed point is unique.*

Proof. To prove that there cannot be two fixed points for a contraction, observe that if $A(x) = x$ and $A(y) = y$, then

$$d(x, y) = d(A(x), A(y)) \leq \alpha d(x, y).$$

Since $\alpha < 1$, this implies that $d(x, y) = 0$ and $x = y$. ■

Another feature of the relation between contractions and fixed points is that if A is a contraction, then the sequence of iterates

$$x_0, A(x_0), A(A(x_0)), A(A(A(x_0))), \ldots$$

must converge to the fixed point or (if there is no fixed point) must diverge.

Lemma 13.73: *Let A be a contraction map, let x_0 be an element of the metric space and construct the sequence*

$$x_1 = A(x_0),\ x_2 = A(x_1),\ x_3 = A(x_2),\ \ldots.$$

Then

1. *If A has a fixed point x, then $x_n \to x$.*
2. *If $x_n \to x$, then x is the fixed point of A.*

Proof. Let s, t be any elements of the metric space. Let A^n denote the composition of A with itself n times. Observe that, by the definition of a contraction, that

$$d(A(s),A(t)) \leq \alpha d(s,t),$$

that

$$d(A^2(s),A^2(t)) \leq \alpha d(A(s),A(t)) \leq \alpha^2 d(s,t)$$

and so, continuing in this way,

$$d(A^n(s),A^n(t)) \leq \alpha^n d(s,t) \tag{9}$$

for all n. Replace s by x_0 and t by x, where x is a fixed point of A so $x = A(x) = A^2(x) = \cdots = A^n(x)$ and obtain

$$d(A^n(x_0),x) \leq \alpha^n d(x_0,x) \to 0 \tag{10}$$

as $n \to \infty$. This proves the first assertion.

To prove the second assertion we first check that every contraction map is continuous (Exercise 13.9.5). Suppose that $x_n \to x$. Note that $x_{n+1} = A(x_n)$ and $x_{n+1} \to x$ and, by continuity, $A((x_n)) \to A(x)$, Thus $A(x) = x$ and x is a fixed point of A. ■

These two lemmas do not establish the existence of a fixed point for a contraction; indeed it is easy to give an example of a contraction with no fixed point. We shall see that a contraction map on a *complete* space must have a fixed point. But observe that a contraction map does more than "move points closer together:" It moves them together by an factor strictly smaller than 1. The following example shows a mapping that is nearly a contraction (on a complete space) but that has no fixed point.

Example 13.74: The function

$$f(x) = x + 1/x$$

maps the complete metric space $[2,\infty)$ into itself, and $f'(x) = 1 - x^{-2}$ satisfies $0 < |f'(x)| < 1$ for all $x \in [2,\infty)$. Thus if x and y are close together, $f(x)$ and $f(y)$ are even closer together. But this function has no fixed point because for all x,

$$x \neq x + 1/x.$$

Notice that it is not, however, a contraction map. The inequality

$$|x + 1/x - (y + 1/y)| \leq \alpha |x - y|$$

cannot hold for all $x, y \in [2,\infty)$ and any $\alpha < 1$ (although it does hold for $\alpha = 1$). [See Figure 13.4. Note that $A/B < 1$ but $\lim_{x_1,x_2\to\infty} A/B = 1$ illustrating that the function f cannot be a contraction map on $[2,\infty)$.] ◄

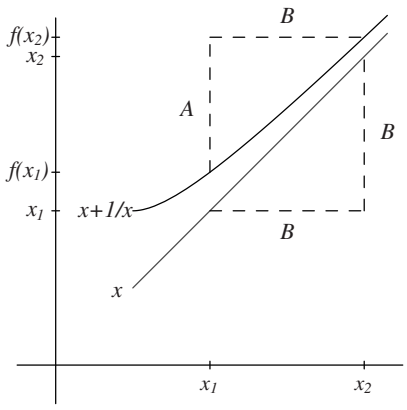


Figure 13.4. Graphs of $f(x) = x + 1/x$ and $g(x) = x$ with $A = d(f(x_1), f(x_2))$ and $B = d(g(x_1), g(x_2))$.

Observe how, in Theorem 13.75, the fact that $\alpha < 1$ for a contraction map plays an essential role in the proof.

Theorem 13.75 (Banach) *A contraction map A defined on a complete metric space (X,d) has a unique fixed point.*

Proof. We obtain the fixed point of A by starting at an arbitrary point x_0 in the space and iterating the function. Let $x_0 \in X$. Let $x_1 = A(x_0)$, $x_2 = A(x_1) = A^2(x_0)$, and, in general,

$$x_n = A(x_{n-1}) = A^n(x_0) \quad (n = 1, 2, \dots).$$

Here we are using the customary notation $A^{n+1}(x) = A(A^n(x))$.

We show that the sequence $\{x_n\}$ is a Cauchy sequence. Let $n \leq m$. Then, using the inequality (9), we obtain

$$d(x_n, x_m) = d(A^n(x_0), A^m(x_0))$$
$$= d(A^n(x_0), A^n(A^{m-n}(x_0))) \leq \alpha^n d(x_0, x_{m-n}).$$

We can estimate the latter term, using the triangle inequality and the inequality (9),

$$d(x_0, x_{m-n}) \leq d(x_0, x_1) + d(x_1, x_2) + \cdots + d(x_{m-n-1}, x_{m-n})$$
$$\leq d(x_0, x_1)[1 + \alpha + \alpha^2 + \cdots + \alpha^{m-n-1}]$$
$$\leq \frac{d(x_0, x_1)}{1 - \alpha}.$$

Let $\varepsilon > 0$. Choose N so that

$$\alpha^N d(x_0, x_1) < \varepsilon(1 - \alpha).$$

Then if $m \geq n \geq N$ we have from these inequalities that

$$d(x_n, x_m) \leq \frac{\alpha^n d(x_0, x_1)}{1 - \alpha} < \varepsilon$$